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MILITARY HANDBOOK

FIRE PROTECTION FOR FACILITIES

ENGINEERING, DESIGN, AND CONSTRUCTION

AMSC N/A

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ABSTRACT

This handbook was produced to provide detailed guidance for the incorporation of fire prevention measures in the design and construction of Department of Defense facililties. Concerns for property, equipment, and personnel were among the comprehensive considerations included in this handbook to ensure safety of human life, minimize injuries, and minimize damage to property and equipment.

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FOREWORD

This handbook has been developed from an evaluation of facilities in the shore establishment, from surveys of the availability of new materials and construction methods, and from selection of the best design practices of the Naval Facilities Engineering Command (NAVFACENGCOM), other Government agencies, and the private sector. This handbook was prepared using, to the maximum extent feasible, national professional society, association, and institute standards. Deviations from this criteria, in the planning, engineering, design, and construction of Naval shore facilities, cannot be made without prior approval of NAVFACENGCOM HQ Code 04.

Design cannot remain static any more than can the functions it serves or the technologies it uses. Accordingly, recommendations for improvement are encouraged and should be furnished to Commander, Naval Facilities Engineering Command, Naval Facilities Engineering Command Headquarters, (Code 04F), 200 Stovall Street, Alexandria, VA 22332-2300; Telephone (703) 325-0270.

THIS HANDBOOK SHALL NOT BE USED AS A REFERENCE DOCUMENT FOR PROCUREMENT OF FACILITIES CONSTRUCTION. IT IS TO BE USED IN THE PURCHASE OF FACILITIES ENGINEERING STUDIES AND DESIGN (FINAL PLANS, SPECIFICATIONS, AND COST ESTIMATES). DO NOT REFERENCE IT IN MILITARY OR FEDERAL SPECIFICATIONS OR OTHER PROCUREMENT DOCUMENTS.

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Section 1: INTRODUCTION

1.1 Scope. The provisions of this handbook are applicable to Department of Defense (DOD) facilities, whether acquired or leased, by appropriated or nonappropriated funds or third party financed/constructed. Provisions of this handbook also apply to major alterations of existing structures and nonpermanent construction as directed. Facilities covered by this document include all types of buildings and their contents, structures (whether considered temporary or permanent), mobile and stationary equipment, waterfront facilities, outside storage, shore protection for ships and craft, and external protection for aircraft. Matters relating to fire department operations, manning, and equipment are not covered by this handbook.

1.2 Cancellation. This handbook cancels and supersedes MIL-HDBK-1008, Fire Protection for Facilities Engineering, Design and Construction, April 1985.

1.3 Criteria. Fire protection criteria shall conform to the requirements of this handbook and of the applicable standards contained in the current National Fire Codes, published by the National Fire Protection Association (NFPA), Loss Prevention Data Sheets, as published by the Factory Mutual (FM) Systems and the Office of Safety and Health Administration (OSHA); and the Uniform Building Code (UBC), published by the International Conference of Building Officials except as modified herein or by specific criteria issued by the appropriate headquarters department. Specific requirements contained in this handbook are necessary to obtain uniform application of fire protection engineering and to highlight problem areas encountered with Department of Defense (DOD) facilities. Special problems not covered by this handbook nor by the NFPA standards nor by applicable department criteria shall be referred to the appropriate department headquarters or DOD agency headquarters for guidance.

Deviation from these criteria, where a valid need exists and where an alternate solution involving equivalent concept and sound fire protection engineering is available, may be accepted after approval by the appropriate headquarters. Request for approval must include justification, hazard analysis, cost comparison, criteria used, and other pertinent data. Lack of funds shall not be considered sufficient justification for deviation below established fire protection standards. Approval granted will apply only to the specific request under consideration and not to cases with similar circumstances.

1.3.1 Air Force. This handbook is used in conjunction with Air Force Regulation (AFR) 88-15, Criteria and Standards for Air Force Construction, when supporting Air Force construction requirements. Where there is additional or expanded guidance in AFR 88-15, the paragraph number is provided in parentheses in the appropriate paragraph.

1.4 Design Analyses. All concept and final design analyses shall address provisions pertaining to fire protection as required by this handbook. Fire protection provisions shall be summarized and submitted as a separate analysis. Where applicable, the following fire protection provisions must be delineated:

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a) Type of construction (including interior finish materials),

b) Classification of occupancy,

c) Building separation or exposure protection,

d) Fire protection criteria (this handbook and pertinent standards and codes),

e) Location of all fire-rated walls including fire-rated doors and dampers with identification as applicable (firewalls, fire partitions, and smoke partitions, with their fire-resistive ratings),

f) Life safety provisions (exit travel distances, exit unit widths based on the capacity and occupant load, number, type, exit signs, and lighting),

g) Automatic extinguishing systems (identification of all sprinklered areas and areas protected by other automatic suppression systems),

h) Water supplies,

i) Smoke control system. Smoke compartments and the requirements for smoke dampers, smoke detectors, and smoke partitions shall be provided. The smoke control system shall be delineated by schematic diagram, when applicable, indicating the operations of the normal HVAC mode and the smoke removal mode.

j) Fire alarm system (the type of alarm system and location of the fire alarm equipment, and fire zones),

k) Fire detection system (the type of detection system and location of detectors, and fire zones),

l) Location of fire extinguisher cabinets and firehose standpipes, and

m) Interior finish ratings.

1.5 Services of Fire Protection Engineers and their Qualifications. All major projects will require the services and review of a fire protection engineering consultant or a registered professional engineer with a fire protection background who has at least 2 years' experience in fire protection design. Qualifications shall be submitted to and approved by the contracting officer.

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Section 2: BUILDING CONSTRUCTION

2.1 Types of Construction

2.1.1 General. Construction shall conform to the fire resistance requirements and fire area limitations of the UBC, except as modified herein. Strategic facilities located in theatres of operation may require special considerations for hardening and survivability which exceed minima set forth in this section.

2.1.2 Definitions

2.1.2.1 Combustible. Combustible materials are those which will ignite, burn, support combustion, or release flammable vapors when subject to heat or fire. The term combustible is not related to any specific ignition temperature or flame-spread rating.

2.1.2.2 Fire Resistance Rating. Fire resistance rating is the time that the construction will withstand the standard fire exposure as determined by the American Society for Testing and Materials (ASTM), ASTM E-119, Standard Methods of Fire Tests of Building Construction and Materials.

2.1.2.3 Flame-Spread Rating. Flame-spread rating is a numerical classification determined by the ASTM E-84, Standard Test Method for Surface Burning Characteristics of Building Materials, which indexes the relative burning behavior of a material by quantifying the spread of flame in a test specimen. The surface burning characteristic of a material is not a measure of resistance to fire exposure.

2.1.2.4 Noncombustible. Noncombustible materials are those which will not ignite, burn, support combustion, or release flammable vapors when subjected to heat or fire.

2.1.2.5 Smoke-Developed Rating. Smoke-developed rating is a numerical classification determined by ASTM-E-84, which indexes the smoke generation rate of a given material to those of two standard materials, asbestos-cement board and select grade red oak.

2.1.2.6 Windowless Structure. Any building or story (including basement) of a building over 1,500 ft^2 (140 m^2) is considered windowless by one of the following definitions:

a) No exterior wall is provided with more than 20 ft^2 (1.85 m^2) of opening entirely above ground level in each 50 linear ft (15.24 m) of wall. Openings shall have a minimum dimension of 30 in. (761.4 mm), and shall be readily accessible to the fire department for fire fighting and rescue operations.

b) Openings as described above are located only on one side of a story, and the opposite wall is more than 75 ft (23 m) from the openings.

2.1.2.7 Underground Structures. A structure in which there is no direct access to outdoors or to another fire area other than by upward travel.

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2.1.3 Construction Categories. Buildings shall be placed in one of five major types of construction categories based primarily upon the fire resistance of interior and exterior structural support members, walls, floors and roofs. Fire resistance ratings of buildings shall conform to the UBC. The construction categories are Types I, II, III, IV and V as defined by the UBC (see Table 1). Table 2 provides a summary of required fire resistances for various building components in each of the categories. Fire resistances are expressed in 1-, 2-, 3- or 4-hour ratings.

2.1.3.1 Sprayed-on Fire Proofing. This method of protecting steel may be ineffective due to practical difficulties in obtaining positive bonds and adequate thickness of materials on all surfaces of the structural steel members. Where this type of system is essential, Type 1 cementitious materials shall be used (refer to NAVFACENGCOM Guide Specification TS-07250, Sprayed Fire Protection for Navy Projects).

2.1.4 Classification of Occupancy. Classification of occupancy is in accordance with the UBC (refer to Appendix A). Buildings shall be classified as either Group A, B, E, H, I, M or R depending upon the existing or proposed life and fire hazard. Major classification groups are described in paras. 2.1.4.1 through 2.1.4.6.

2.1.4.1 Group A Occupancies. Places of assembly which include churches, chapels, clubs (officer, enlisted personnel, etc.), drill halls, gymnasiums, schools and stadiums.

2.1.4.2 Group B Occupancies. Gasoline service stations, garages, and aircraft hangars where no repair work is done except the exchange of parts and maintenance requiring no open flame, welding or the use of highly flammable liquids; exchanges, commissaries, office buildings, printing plants, factories, and workshops using materials not highly flammable.

2.1.4.3 Group E Occupancies. Educational and dependent-care facilities.

2.1.4.4 Group H Occupancies. Woodworking shops, aircraft repair hangars, vehicle repair garages and other facilities where hazardous and highly flammable materials are used, stored or dispensed.

2.1.4.5 Group I Occupancies. Nurseries, hospitals, medical and dental facilities housing nonambulatory patients; detention and correctional facilities.

2.1.4.6 Group M Occupancies. Vehicle parking facilities, sheds, agricultural buildings.

2.1.4.7 Group R Occupancies. Barracks, family, and bachelor housing.

2.1.5 Types of Construction for Various Occupancies

2.1.5.1 Fire Resistive Construction (Types I and II Fire Resistive). The following will be of fire resistive construction:

a) Hospitals over one story in height.

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Table 1
Uniform Building Code and NFPA 220 Equivalents

UBC		NFPA 220
Type I (F.R.)	=	Type 1 (444) Type I (332)
Type II (F.R.)	=	Type II (222)
Type II (1 – hour)	=	Type II (111)
Type II (N)	=	Type II (000)
Type III (1 – hour)	=	Type III (211)
Type III (N)	=	Type III (200)
Type IV (H.T.)	=	Type IV (2HH)
Type V (1 – hour)	=	Type V (111)
Type V (N)	=	Type V (000)

b) Barracks, bachelor quarters, and administrative facilities exceeding three stories in height. Fire areas per floor for barracks and bachelor quarters in this type of construction are not limited.

c) Dependent school buildings more than three stories in height.

d) Buildings or areas within buildings, such as vaults, which are required to prevent outside fires from destroying or damaging the contents.

e) Buildings or areas within buildings used for the storage of hazardous chemicals or materials in order that a fire within the building or within the space of the building will be prevented from spreading to the adjoining building(s) or areas.

2.1.5.2 Protected Noncombustible Construction (Type II - 1 Hour). This category includes one-story hospitals and all other buildings exceeding three stories in height, and not included in para. 2.1.5.1.

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Table 2
Types of Construction--Fire-Resistive Requirements[1]
(In Hours)

BUILDING ELEMENT	TYPE I / TYPE II NONCOMBUSTIBLE				TYPE III / TYPE IV / TYPE V COMBUSTIBLE				
	FIRE RESISTIVE	FIRE RESISTIVE	1 – HR	N	1 – HR	N	HT	1 – HR	N
Exterior Bearing Walls	4	4	1	N	4	4	4	1	N
Interior Bearing Walls	3	2	1	N	1	N	1	1	N
Exterior Nonbearing Walls	4	4	1	N	4	4	4	1	N
Structural Frame[2]	3	2	1	N	1	N	1 or HT	1	N
Partitions – Permanent[3]	1	1	1	N	1	N	1 or HT	1	N
Shaft Enclosures	2	2	1	1	1	1	1	1	N
Floors	2	2	1	N	1	N	HT	1	1
Roofs	2	1	1	N	1	N	HT	1	N

N – No general requirements for fire resistance. HT – Heavy Timber

[1] Adapted from Uniform Building Code, 1985 Edition, Copyright 1982, with permission of the publisher, the International Conference of Building Officials.

[2] Structural frame elements in the exterior wall shall be protected against external fire exposure as required for exterior bearing walls or the structural frame, whichever is greater.

[3] A permanent partition is usually of masonry construction and is not intended to be altered or changed.

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2.1.5.3 Unprotected Noncombustible Construction (Type II N). Buildings, except as stated in paras. 2.1.5.1, 2.1.5.2 and 2.1.5.4, shall be of unprotected noncombustible construction. Dependent school buildings two and three stories in height shall be of this type of construction. Bents of glued laminated wood may be used in this type of construction. Fire areas for barracks and bachelor quarters in this type of construction shall be limited to 20,000 ft^2 (1858 m^2) per floor.

2.1.5.4 Combustible Construction (Types III or Type V). Small and light hazard buildings, not specifically covered herein, may be of combustible construction where warranted by local conditions or where economies may be effected by the use of locally available materials and construction methods.

2.2 Fire Areas and Height Limitations

2.2.1 Fire Areas. Fire areas (the total area bounded by firewalls) shall conform with the UBC except as modified herein. Maximum allowable fire areas for given occupancy categories and construction types are summarized in Table 3. Fire areas may be increased in accordance with the requirements set forth in the UBC. Facilities used for shippping, receiving, packing and processing of materials are considered as warehouses. Exceptions are as given in paras. 2.2.1.1 to 2.2.1.4.

2.2.1.1 General Warehouses. General warehouses shall have fire areas not exceeding 40,000 ft^2 (3703 m^2). Fire areas in warehouses should be separated by walls which have 4-hour fire resistance ratings.

2.2.1.2 Depot Level Warehouses. For depot level warehouses, bay sizes may be increased up to 120,000 ft^2 (7407 m^2) between 4-hour firewalls upon a finding and determination by the authority having jurisdiction that all the following conditions are met:

a) The increased size is required for efficient operation.

b) The possible additional loss has been recognized and is acceptable.

c) Other fire safety features (such as draft curtains, vents, and large-drop fast-response sprinkler heads) have been designed to compensate for the additional hazard insofar as such design is practicable.

2.2.1.3 Hazardous Materials Storage. Warehouses storing hazardous materials shall comply with the applicable NFPA standard and in no case exceed 20,000 ft^2 (1852 m^2) between 4-hour firewalls.

2.2.1.4 Unsprinklered Attics and Suspended Ceilings. Draft stops of gypsum board on wood or metal frame are required in unsprinklered attic spaces of combustible roof construction or in unsprinklered combustible suspended ceiling spaces to divide the spaces into areas not exceeding 3,000 ft^2 (280 m^2). Self-closing and latching access doors of similar construction shall be provided in the draft stop where there is no other means of access to an area.

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Table 3
Basic Allowable Floor Area for Buildings One Story in Height[1]
(In Ft[2])

OCCUPANCY[2]	TYPES OF CONSTRUCTION								
	I		II		III		IV	V	
	FR	FR	ONE – HOUR	N	ONE – HOUR	N	HT	ONE – HOUR	N
A – 1	Unlimited	29,900	Not Permitted						
A)2 – 2.1	Unlimited	29,900	13,500	Not Permitted	13,500	Not Permitted	13,500	10,500	Not Permitted
A)3 – 4	Unlimited	29,900	13,500	9,100	13,500	9,100	13,500	10,500	6,000
B)1 – 2 – 3	Unlimited	39,900	18,000	12,000	18,000	12,000	18,000	14,000	8,000
B – 4	Unlimited	59,900	27,000	18,000	27,000	18,000	27,000	21,000	2,000
E	Unlimited	45,200	20,200	13,500	20,200	13,500	20,200	15,700	9,100
H)1 – 2	15,000	12,400	5,600	3,700	5,600	3,700	5,600	4,400	2,500
H)3 – 4 – 5	Unlimited	24,800	11,200	7,500	11,200	7,500	11,200	8,800	5,100
I)1 – 2	Unlimited	15.100	6,800	Not Permitted	6,800	Not Permitted	6,800	5,200	Not Permitted
I – 3	Unlimited	15.100	Not Permitted						
M	Refer to Chapter 11 of the Uniform Building Code								
R – 1	Unlimited	29,900	13,500	9,100	13,500	9,100	13,500	10,500	6,000
R – 3	Unlimited								

N – No requirements for fire resistence
FR – Fire Resistive
HT – Heavy Timber

NOTE: For H – 6 occupancies (Semiconductor Production Facilities), see the latest edition of UBC

NOTE: Fire areas may be increased when automatic sprinklers are installed and/or when adequate separation distances are provided

[1] Adapted from Table No. 5 – C of the Uniform Building Code, 1985. Edition, Copyright 1982, with permission of the publisher, the International Conference of Building Officials.

[2] Refer to para 2.1.4 and Appendix A.

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2.2.2 Building Height Limitations. Building height limitations shall conform to the UBC except as modified by specific criteria issued by the Military Departments. Maximum building heights shall not exceed that given for the type of construction and occupancy in Table 4. Building heights may be increased when automatic sprinkler protection is provided per UBC.

2.3 Separation of Structures

2.3.1 Exposure Classification. The exposure classification, given in Table 5, is a function of the occupancy's hazard classification and is used for determining separations. Occupancy hazard classifications are: extra hazard, ordinary hazard (Groups 1, 2 and 3) and light hazard. These are defined and examples provided in Appendix B of this handbook.

2.3.2 Separation Tables. Separation requirements listed herein are applicable only when combined fire areas exceed the fire area limitations outlined in para. 2.2 of this handbook. The required clearances between structures shall be determined by applying the following rules and Tables 5 and 6. First, Table 5 is used to determine the exposure classification of the exposing building. That classification A, B, C, or D is then used in Table 6 to determine a separation distance based upon the exposed wall surface of the exposed building. Separation distances are governed by the building having the greater hazard, not by which building is existing and which is planned. Therefore, in some cases it will be necessary to determine separation distances for both buildings and to choose the larger.

2.3.3 Rules for the Use of Tables 5 and 6. The following rules shall apply to Tables 5 and 6:

a) Rule 1: When the existing building has no automatic sprinklers, and the new building has automatic sprinklers, Table 5 shall be used for the existing building to determine a classification of exposure, and then Table 6 shall be used for new building to determine required separation distance.

b) Rule 2: When the existing building has automatic sprinklers and the new building has no automatic sprinklers, Table 5 shall be used for the new building to determine a classification of exposure, and then Table 6 shall be used also for the new building to determine required separation distances.

c) Rule 3: When both buildings have automatic sprinklers and at least one building has a 2-hr or greater masonry-parapeted firewall, no separation shall be required.

d) Rule 4: When both buildings have automatic sprinklers and neither has a 2-hr or greater masonry-parapeted firewall, each building shall be considered the exposing building to determine the maximum exposure classification from Table 5. Using that exposure, determine separation from Table 6 for blank masonry.

e) Rule 5: When neither building has automatic sprinklers, determine separation distances twice, considering each building as the exposing building. Use the larger of the determined separation distances as the minimum requirement.

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Table 4
Maximum Height of Buildings Types of Construction[1]

OCCUPANCY[2]	TYPES OF CONSTRUCTION								
	I	II			III		IV	V	
	FR	FR	ONE – HOUR	N	ONE – HOUR	N	HT	ONE – HOUR	N
	Maximum Height In Feet								
	Unlimited	160	65	55	65	55	65	50	40
	Maximum Height In Stories								
A – 1	Unlimited	4				Not Permitted			
A)2 – 2.1	Unlimited	4	2	Not Permitted	2	Not Permitted	2	2	Not Permitted
A)3 – 4	Unlimited	12	2	1	2	1	2	2	1
B)1 – 2 – 3	Unlimited	12	4	2	4	2	4	3	2
B – 4	Unlimited	12	4	2	4	2	4	3	2
E	Unlimited	4	2[3]	1	2[3]	1	2[3]	2[3]	1
H – 1	Unlimited	2	1	1	1	1	1	1	1
H – 2 – 3 – 4 – 5	Unlimited	5	2	1	2	2	2	2	1
I – 1	Unlimited	3	1	Not Permitted	1	Not Permitted	1	1	Not Permitted
I – 2	Unlimited	3	2	Not Permitted	2	Not Permitted	2	2	Not Permitted
I – 3	Unlimited	2			Not Permitted				
M	Refer to Chapter 11 of the Uniform Building Code								
R – 1	Unlimited	12	4	2	4	2	4	3	2
R – 3	Unlimited	3	3	3	3	3	3	3	3

N – No requirements for fire resistence
FR – Fire Resistive
HT – Heavy Timber

NOTE: Fire areas may be increased when automatic sprinklers are installed and/or when adequate separation distances are provided.

1 Adapted from Table No. 5 – D of the Uniform Building Code, 1985 Edition, Copyright 1982, with permission of the publisher, the International Conference of Building Officials.

2 Refer to para 2.1.4 and Appendix A.

3 Kindergarten, day care, 1st and 2nd grade occupancies shall not be located above the first story.

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Table 5
Classification of Exposure Fire

CONSTRUCTION OF EXPOSING BUILDING			HAZARD DUE TO OCCUPANCY	EXPOSURE CLASSIFICATION FIRE AREA Ft2 (m 2) PER FLOOR				
HEIGHT	WALLS	ROOF & FLOOR		1,500 – 2,500 (140 – 230)	2,500 – 5,000 (230 – 460)	5,000 – 10,000 (460 – 930)	10,000 – 20,000 (930 – 1,860)	Over 20,000 (1,860)
Multi – Story	Load Bearing Masonry	Comb	Extra	C	B	B	B	A
			Ordinary 2,3	C	C	B	B	B
			Ordinary 1	D	D	C	C	B
			Light	D	D	D	C	B
One Story[3]	Load Bearing Masonry	Comb	Extra	C	B	B	B	B
			Ordinary 2,3	C	C	B	B	B
			Ordinary 1	D	D	C	C	B
			Light	D	D	D	D	C
Multi – Story	Comb[4]	NC or Comb	Extra	B	B	A	A	A
			Ordinary 2,3	B	B	B	B	A
			Ordinary 1	B	B		B	B
			Light	C	B	B	B	B
One Story[3]	Comb[4]	NC or Comb	Extra	B	B	B	B	A
			Ordinary 2,3	C	B	B	B	A
			Ordinary 1	C	B	B	B	B
			Light	D	C	C	B	B
Multi – Story	NC Panel[5] on Steel Frame Non – Load – Bearing Masonry, Concrete or Brick	NC or Comb	Extra	C	B	B	B	B
			Ordinary 2,3	C	C	C	B	B
			Ordinary 1	D	D	D	C	C
			Light	D	D	D	D	C

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Table 5 (Continued)
Classification of Exposure Fire

CONSTRUCTION OF EXPOSING BUILDING			HAZARD DUE TO OCCUPANCY [2]	EXPOSURE CLASSIFICATION FIRE AREA Ft2 (m 2) PER FLOOR				
HEIGHT	WALLS	ROOF & FLOOR [1]		1,500 – 2,500 (140 – 230)	2,500 – 5,000 (230 – 460)	5,000 – 10,000 (460 – 930)	10,000 – 20,000 (930 – 1,860)	Over 20,000 (1,860)
One Story [3]	NC Panel [5] on Steel Frame Non – Load – Bearing Masonry, Concrete or Brick	NC or Comb	Extra	C	C	B	B	B
			Ordinary 2,3	C	C	B	B	B
			Ordinary 1	D	D	D	D	D
			Light	D	D	D	D	D
Single or Multi – Story	Load – Bearing Masonry or Concrete	NC	Extra	D	D	C	C	C
			Ordinary 2,3	D	D	C	C	C
			Ordinary 1	D	D	D	D	D
			Light	D	D	D	D	D

Definitions: Comb – Combustible
NC – Noncombustible.
Examples of Comb: Plank on timber; plank on steel; board on joist.

[1] If floors and roof are of different construction, use worst case.

[2] Based on occupancy classifications as defined in Appendix B.

[3] One – story buildings 25 ft (8m) or higher shall be considered as multistory

[4] Includes old type asphalt coated metal buildings.

[5] NC panel wall may be protected metal, plain or corrugated metal with or without insulation, asbestos – cement, or similar NC materials on steel frame.

Notes: Consider fire walls, if any, when determining fire area. For explosive ordnance materials, refer to DOD 5154.4S – DOD Ammunition and Explosive Safety Standards.

This table and Table 6 were adapted from FM, Loss Prevention Data Sheets, based upon theoretical radiation calculations.

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Table 6
Separation Distances in Ft (m)

EXPOSED BUILDING (WALL SURFACE)	EXPOSURE FROM TABLE 5			
	A	B	C	D
Wood or Wood Frame				
Asphalt shingle	150 (46)	110 (33)	90 (27)	70 (21)
Painted or unpainted	95 (29)	65 (20)	50 (15)	40 (12)
Steel or aluminum siding	95 (29)	65 (20)	50 (15)	40 (12)
Stucco or asbestos shingle	75 (23)	45 (14)	35 (11)	25 (8)
Veneer of brick or stone	60 (18)	40 (12)	25 (8)	15 (5)
Panel on Steel Frame[1]				
Insulated aluminum, plastic, or fiberglass panel	95 (29)	70 (21)	50 (15)	35 (11)
Asphalt – coated metal	75 (23)	45 (14)	35 (11)	25 (8)
Insulated steel, panel	65 (20)	40 (12)	30 (9)	20 (6)
Protected metal, plain metal	55 (17)	35 (11)	25 (8)	15 (5)
Asbestos panel	45 (14)	30 (9)	20 (6)	10 (3)
Masonry or Concrete				
Blank	20 (6)	15 (5)	0	0
Wired glass and window sprinklers	30 (9)	25 (8)	15 (5)	0
Wired glass	40 (12)	35 (11)	20 (6)	10 (3)
Plain glass	60 (18)	40 (12)	25 (8)	15 (5)

[1] Wired glass shall be installed in window openings when the separation distance is within 60 ft (18 m) for Class A exposure, 40 ft (12 m) for Class B exposure, 25 ft (8 m) for Class C exposure, and 10 ft (3 m) for Class D exposure.

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f) Rule 6: When either building has a 2-hr or greater fire-rated wall but no automatic sprinklers, consider this building to be the exposing building when determining the exposure classification. Determine the classification of exposure from Table 5. Next, use Table 6 for the other building to determine required separation distance.

g) Rule 7: When the taller building has a 4-hr or greater masonry free-standing parapeted blank fire-rated wall, no separation is required.

h) Rule 8: When the taller building has automatic sprinklers and a 2-hr or greater masonry parapeted blank fire-rated wall, no separation is required.

i) Rule 9: When both buildings have 2-hr or greater masonry parapeted blank fire-rated walls, even though they are not free standing, no separation is required.

j) Rule 10: For buildings having less than 1,500 ft^2 (139.35 m^2) of fire area, go directly to Table 6 and use Classification D.

2.3.4 Parapets. Parapets are not required where roof decks are of heavy noncombustible constructon such as concrete or gypsum. For all other roof decks, parapets shall be 30 in. (764 mm). Where roof decks are FM approved, Class I steel deck, a 12-in. (306 mm) parapet is required. The 30-in. or 12-in. dimension is measured from the top surface of the lower roof to the top of the parapet.

2.3.5 Roof Exposure. Exposed building roof covering shall be FM approved or classified by Underwriters' Laboratories (UL) as Class A. Exposed roof covering not meeting these requirements shall be graveled, 3 to 4 lb/ft^2 (15 to 20 kg/m^2), in a flood coat of tar or asphalt. It is necessary to place the gravel within the safe separation distance given in Table 6 for asphalt shingle.

2.3.6 Special Requirements. Certain structures must meet special requirements as listed below.

a) Family housing separation requirements shall conform to HUD Minimum Property Standards.

b) House trailers shall be separated in accordance with National Fire Protection Association (NFPA) Standard NFPA No. 501A, Mobile Homes Including Mobile Home Park Requirements.

c) Relocatable or prefabricated structures used for other than family dwellings shall be separated in accordance with requirements for permanent buildings.

d) Relocatable facilities such as electronic equipment vans shall not be grouped to form areas greater than 6,000 ft^2 (557.4 m^2) for unsprinklered facilities and 12,000 ft^2 (1114.8 m^2) for sprinklered facilities. Relocatable facilities having extra hazardous occupancies (refer to Appendix B, para. 1.5) shall not be grouped to form areas greater than 4,000 ft^2 (372 m^2) for unsprinklered facilities and 8,000 ft^2 (743 m^2)

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for sprinklered facilities. A minimum separation of 50 ft (15 m) shall be provided for extra-hazardous groups and 15 ft (5 m) for all other groups.

2.4 Limiting Interior Fire Spread

2.4.1 Vertical Fire Cutoffs. Enclosures for stairways, elevators, ducts, chases, etc. shall be in accordance with the following:

a)	NFPA 80	Fire Doors and Windows
b)	NFPA 90A	Standard for the Installation of Air Conditioning and Ventilating Systems
c)	NFPA 90B	Standard for the Installation of Warm Air Heating and Air Conditioning Systems
d)	NFPA 91	Standard for the Installation of Blower and Exhaust Systems for Dust, Stack and Vapor Removal or Conveying
e)	NFPA 96	Vapor Removal for Cooking Equipment
f)	NFPA 101	Life Safety Code
g)	NFPA 211	Standard for Chimneys, Fireplaces, and Vents
h)	NFPA 220	Building Construction, Standard Types

Where cables, conduit, telephone wire, ducts, etc., penetrate floors or firewalls, fire stops shall be provided. Fire stops shall be by an assembly tested in accordance with ASTM E-814 and shall have a rating equal to the fire resistance rating of the wall or floor that is penetrated.

2.4.2 Protection of Openings. Door openings shall be protected in accordance with NFPA 80, Fire Doors and Windows. Fire door assemblies are required on each side of the door opening in 4-hr firewalls when openings are fitted with rolling or sliding doors for material handling. Personnel type doors in such walls shall be single 3-hr fire doors. All fire doors shall be listed by UL, or approved by FM laboratories. Local fabrication of fire doors shall not be permitted. Protection of conveyor openings through firewalls shall be in accordance with NFPA 80. The protection of ducts at point of passage through firewalls shall be in accordance with NFPA 90A, Air Conditioning and Ventilating Systems and/or NFPA 90B, Warm Air Heating and Air Conditioning Systems.

2.5 Exits

2.5.1 Requirements. All structures shall have exit facilities and means of egress that comply with the requirements of NFPA 101, Life Safety Code, except as modified by this handbook.

2.5.1.1 Radioluminous Exit Signs. Various DoD radiation safety committees are reviewing the radiologic aspects of these signs. Certain applications (such as where electrical power is not available, where long distances of wire and conduit make installation of standard exit signs impractical or where explosion proof signs are required) may be appropriate. There is no requirement to remove existing installations. Prior to the installation of such devices, the appropriate military headquarter command or DoD agency headquarters must be contacted for approval.

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2.6 Interior Finish

2.6.1 General. Interior finish is defined as the material of walls, ceilings, wainscoating, and other interior surfaces of a building, and other interior surfacing materials applied to the walls, movable partitions, and ceilings. Exposed insulating and acoustical materials are considered in the category of interior finish.

2.6.2 Requirements. Specific instances associated with structures are presented in paras. 2.6.2.1 through 2.6.2.3.

2.6.2.1 New Construction, Alterations, and Rehabilitation. This para. cross references para. 1-57 of AFR 88-15, Criteria and Standards for Air Force Construction. Wall and ceiling finishes and movable partitions shall conform to requirements of NFPA 101, Life Safety Code, except as follows:

a) Interior finish for all exits, hospital patient rooms, and sleeping rooms, shall be Class A only. Interior finish for all other areas shall be either Class A or Class B. Class B finish is permitted in hospital patient rooms of not over four persons in capacity and in individual rooms of not over four persons in capacity and in individual rooms where automatic sprinkler protection is provided.

b) Use of Class C, D, and E materials is not permitted.

c) Smoke-developed ratings by ASTM E-84, Standard Test Method for Surface Burning Characteristics of Building Materials, test shall not exceed 50 for Class A materials and 100 for Class B materials.

d) Cellular plastics shall not be used as interior finish materials. Drop ceiling (foam-grid panels) may be used in existing buildings if they are listed by UL or approved by Factory Mutual for installation under automatic sprinkler systems. They must be installed in strict accordance with UL or FM instructions and automatic sprinklers must be installed above the panels.

e) For incidental trim, refer to NFPA 101.

2.6.2.2 Existing construction. In the case of combustible interior finishes in existing buildings, certain basic safeguards are essential. Several alternate measures are available to provide the necessary safeguards to protect life and property against fire, and are as follows:

a) Cover combustible surfaces with gypsum board or other materials meeting the requirements of para. 2.6.2.1.

b) Protect the building with an automatic sprinkler system.

c) Coat all exposed combustible surfaces with an UL-listed fire retardant paint or coating.

2.6.2.3 Carpet Systems and Underlay. Flammability and radiant flux standards apply to carpets and their underlay. Federal law requires that all carpet manufactured for sale shall comply with CPSC-16 CFR 1630, Standard for the Surface Flammability of Carpets and Rugs.

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Carpet systems for corridors shall meet specific acceptance criteria when tested in accordance with ASTM E-648, Critical Radiant Flux of Floor Covering Systems Using a Radiant Flux Heat Source. For this purpose, a corridor is defined as an enclosed space connecting a room or compartment with an exit and may include normal extensions such as lobbies and other enlarged areas.

a) Flammability Requirements for Corridor Carpet Systems. A flame propagation index of less than 4.0 as determined by CPSC-16 CFR 1630, is acceptable for corridor systems in all facilities.

b) Radiant Flux Requirements for Corridor Carpet Systems. The minimum acceptance criteria (average critical radiant flux) as determined by ASTM E-648 for corridor carpet systems and other interior floor finishes, shall be as follows:

(1) 0.50 watts per square centimeter or greater for bachelor housing, hospitals, temporary lodging facilities, and child care facilities.

(2) 0.25 watts per square centimeter or greater for carpet systems in corridors of all other facilities and in electronic equipment areas.

2.7 Insulation

2.7.1 Requirements. Thermal and acoustical insulation shall have a flame-spread rating not higher than 75, and a smoke-developed rating not higher than 150 by ASTM E-84, Standard Test Method for Surface Burning Characteristics of Building Materials, test. Cellular plastic insulation shall be tested in the same densities and thicknesses as the material that will be used in construction applications.

2.7.2 Exceptions. For certain special types of installations the exceptions described in paras. 2.7.2.1 and 2.7.2.2 shall apply.

2.7.2.1 Flame-Spread 100 - No Smoke Limitation. Compliance with the smoke-developed limitation is not required, and a greater flame-spread rating up to 100 is permitted for insulation, including insulating sheathing installed within wall assemblies. In such installations, the interior finish materials shall conform to para. 2.6 of this handbook and shall have a minimum fire-retardant rating of 15 minutes by ASTM E-119 Test, Standard Methods of Fire Tests of Building Construction and Materials, when tested as it will be installed in the actual construction application.

2.7.2.2 No Flame-Spread or Smoke Limitation. Compliance with flame-spread and smoke-developed limitations are not required for the following applications:

a) Insulation installed above poured-concrete or poured-gypsum roof decks, nominal 2-in. (50.8 mm) thick tongue-and-groove wood plank roof decks, or precast roof deck panels or planks that are approved by FM as noncombustible roof deck construction.

b) Insulation installed above roof decks where the entire roof construction assembly, including the insulation, is UL listed as Fire-Classified or FM approved for Class I roof deck construction.

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d) Insulation contained entirely within panels where the entire panel assembly used in the construction application meets the cited flame-spread and smoke-developed limitations.

e) Insulation isolated from interior of building by masonry walls, masonry cavity walls, insulation encased in masonry cores, or concrete floors.

f) Insulation installed over concrete floor slab and completely covered by wood T&G flooring without creating air spaces within the flooring system.

g) Insulation completely enclosed in hollow metal doors.

h) Insulation installed between new exterior siding materials and existing exterior siding or wood board, plywood, fiberboard, or gypsum exterior wall sheathing.

NOTE: The exception to smoke-developed limitation in para. 2.7.2.2 is not applicable to hospitals and correctional facilities.

2.8 Roof Coverings and Roof Deck Assemblies

2.8.1 Roof Coverings. (This para. cross references para. 7-5 of AFR 88-15, Criteria and Standards for Air Force Construction.) All roof coverings shall be approved and listed by UL. The UL Building Materials Directory lists three classes (A, B, and C) of acceptable roof coverings based upon UL 790, Test Methods for Fire Resistance of Roof Covering Materials. Class C roof coverings shall be restricted to housing and small, insignificant buildings over 8,000 ft^2 (2,438.4 m^2) with light exposure.

2.8.2 Roof Deck Assemblies. Assemblies acceptable from an interior fire exposure standpoint are listed as Class I in the FM, Approval Guide and as Fire-Classified in UL, Building Materials Directory. Roof deck assemblies shall be of acceptable type when used in buildings that are not fully sprinklered.

2.9 Alterations, Rehabilitation, and New Construction

2.9.1 Requirements. Adequate fire protection measures shall be incorporated in all alteration, rehabilitation, repair and new construction projects, and shall be taken consistent with the requirements of this handbook.

2.9.2 Fire Protection During Construction. The contract specifications shall refer to the Army Corps of Engineers Manual EM 385-1-1, Safety and Health Requirements Manual, and NFPA 241, Building Construction and Demolition Operations, and shall contain the requirement that the activity fire regulations shall be followed.

2.10 Access for Firefighting

2.10.1 Requirements. A minimum of two stairways or permanently attached noncombustible ladders shall extend to the roof to provide access for fire-fighting in buildings three stories or higher. Access for firefighting shall be provided in one story, windowless buildings by doors, and in two- or more

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story windowless buildings by stair towers at intervals of not over 150 ft (46 m). Underground structures shall also be provided with access for firefighting purposes.

2.11 Air Handling

2.11.1 General. Fire doors or fire dampers shall be installed in ducts which penetrate firewalls, fire partitions and floors.

2.11.2 Design Requirements. Air handling, ventilation, and exhaust system shall meet the design requirements of the following, except as modified by this handbook:

a) NFPA 75 Protection of Electronic Computer/Data Processing Equipment
b) NFPA 90A Air Conditioning and Ventilating Systems
c) NFPA 90B Warm Air Heating and Air Conditioning Systems
d) NFPA 91 Blower and Exhaust Systems
e) NFPA 96 Commercial Cooking Equipment, Vapor Removal
f) NFPA 101 Life Safety Code

2.11.2.1 Corridors. Corridors shall not be used in lieu of ducts for air handling (either supply or return) in hospitals, dispensaries, bachelor quarters, temporary lodging facilities, schools, places of assembly, and in windowless buildings. Use of corridors for air plenums shall be avoided in other occupancies.

2.11.2.2 Plenums. Plenums may be used as an integral part of an air handling system only if they conform to the requirements of NFPA 90A, Air Conditioning and Ventilating Systems. Under no circumstances may combustible materials be located within the air plenum space. All electrical wiring passing through the space, including telephone and communication wiring, shall be approved for that type of environment or shall be in metal conduit. Rooms or areas which form a plenum space or which are used as a plenum shall not be occupied for any purpose except during repairs or maintenance operations to the air handling equipment.

2.11.2.3 Duct Smoke Detectors. Duct smoke detectors are not required in air handling units of less than 15,000 cfm (424.5 m^3/min) capacity, or in units serving one fire area.

Exception: Duct smoke detectors are required in air handling units under 15,000 cfm (424.5 m^3/min) capacity which are located in hospitals, schools, and detention facilities which are not sprinklered.

2.11.3 Smoke and Heat Vents. UL listed smoke and heat vents shall be provided in roofs of buildings containing materials having high heat-release potential, such as warehouses, flammable liquid storage and handling facilities, and other extra hazard occupancies. NFPA 204, Smoke and Heat Venting Guide, provides guidance.

2.12 Plastic Pipe and Conduit

2.12.1 Requirements. Plastic piping and conduit used in building construction shall adhere to the following requirements:

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a) Electrical installations shall comply with NFPA 70.

b) Vertical distribution in multiple story buildings shall be made with the piping or conduit installed in fire rated shafts. Shafts shall be a minimum of 1-hour fire rated construction.

c) Piping and conduit which penetrate firewalls or fire partitions shall be metallic and the metallic piping shall extend 6 in. (150 mm) beyond the wall on each side. In lieu of metallic penetrations, approved fire stopping devices may be used. In all cases, penetrations shall maintain the fire resistance integrity when tested in accordance with ASTM E-814, Fire Tests of Through-Penetration Fire Stops.

d) Plastic pipe or conduit shall not be installed in air plenums unless they are approved or listed for that application.

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Section 3: COMMON HAZARDS

3.1 Heating Equipment

3.1.1 Requirements. The appropriate following standard(s) shall be followed:

a)	NFPA 31	Installation of Oil Burning Equipment
b)	NFPA 54	National Fuel Gas Code
c)	NFPA 58	Storage and Handling of Liquefied Petroleum Gases
d)	NFPA 85F	Installation and Operation of Pulverized Fuel Systems
e)	NFPA 85	Oil and Natural Gas-Fired Single Burner Boiler-Furnaces
f)	NFPA 85B	Natural Gas-Fired Multiple Burner Boiler-Furnaces
g)	NFPA 85D	Prevention of Furnace Explosions in Fuel Oil-Fired Multiple Burner Boiler-Furnaces
h)	NFPA 85E	Prevention of Furnace Explosions in Pulverized Coal-Fired Multiple Burner Boiler-Furnaces
i)	NFPA 89M	Clearances for Heat Producing Appliances
j)	NFPA 211	Chimneys, Fireplaces and Vents

3.1.2 Gas. Gas service mains shall not be permitted within the perimeter of foundation lines and shall be installed in accordance with NFPA 54. Natural draft cross ventilation shall be provided for building crawl spaces containing gas service piping. Supply connections from the gas service mains shall rise above grade outside the foundation wall and pass through a full swing joint or a loop of metallic tubing before entering the building. This will avoid pipe rupture in the event of differential settlement or earthquake. Pressure regulators shall be located outside of buildings or vented to outside. Standards for heating system components common to all fuel systems are provided in NFPA 211. Related information is available also in the UBC. Heating system devices shall be approved by a recognized testing laboratory such as UL, FM, or American Gas Association.

3.1.3 Coal

3.1.3.1 Storage. Coal storage shall comply with the following restrictions:

a) Do not store coal in contact with any source of external heat, such as piping, flues, boiler walls, and over steam mains, even though buried. Also avoid storage over iron or steel fire service water mains.

b) Coal shall not be piled over manhole covers or covered pipe trenches that might allow air to find its way into the pipe. Piles of coal shall not be arranged around or be in contact with timbers, columns, or large pipes, as air may pass along these surfaces and produce a flue effect. Coal piles shall not be vented with pipes or flues, because this usually supplies more oxygen and increases the heating.

c) Low-grade coal shall not be piled higher than 10 ft (3 m) and best-grade not higher than 15 ft (4.5 m), unless it is piled by the roll-packing method. Locate yard piles at least 25 ft (7.6 m) from other

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combustibles and from important structures.

d) Coal bins, silos, or bunkers shall be constructed entirely of noncombustible material, preferably concrete. The structure should be roofed over to keep out rain and snow, and the space above the coal sufficiently ventilated to prevent the accumulation of gases given off by the coal. An elevated cone-shaped bin in which coal is fed at the top and removed at the bottom is recommended. This arrangement prevents fine material from collecting and remaining in the bin for long periods. The coal at the bottom, which is most likely to be troublesome, is the first to be removed. Coal bins, and when possible, bunkers and silos, should be emptied during summer shutdowns and other prolonged idle periods. Provide automatic sprinkler protection if the storage facility has combustible construction or occupancy other than coal. Provide access openings for manual firefighting.

3.1.3.2 Handling. Where combustible conveyor belts are used to transport coal, the following shall be observed:

a) An automatic sprinkler system shall be provided. It shall be hydraulically designed to operate ten automatic sprinklers and two small hose streams. Sprinkler spacing should be 100 ft^2 (9.3 m^2) per head. The system should be designed for a flowing pressure of 10 psi (70 kPa) on the end sprinkler, which will provide approximately 200 gpm (12.6 l/s) for sprinklers and hose streams. (In a conveyor enclosure less than 15 ft (4.6 m) wide, this can usually be accomplished by installing a single continuous supply line with an unlimited number of sprinklers.) Water supply should be adequate for at least 1 hr duration. System should be interlocked with the belt drive to shut down on sprinkler water flow.

b) Provide either small hose lines, FM approved 1-1/2 in. (38.1 mm), or hydrants at suitable intervals such that the entire belt is accessible for firefighting.

c) Provide each conveyor belt system with tamperproof devices arranged to automatically shut off driving power in the event of greater than 20 percent belt slowdown or misalignment of belts. Interlocking devices should also shut off power to all contributing conveyors.

3.1.3.3 Pulverizing Equipment. Considerations for coal pulverizing equipment are as follows:

a) All equipment which could contain pulverized coal and air mixtures between the inlet of the raw fuel feeder and the point of pulverized fuel delivery to a firebox should be designed to withstand an internal pressure of 50 psi (345 kPa), after allowing for expected corrosion and deterioration. Tensile stresses should not exceed 1/4 of the ultimate strength, or 5/8 of the yield strength of the material.

NOTE: Exceptions are portions of the system which have adequate explosion venting, areas protected by explosion suppression systems, or areas operating under inert atmospheres.

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b) Electrical equipment in pulverizer areas should be suitable for Class II (NEC designation) hazardous areas. In normally clean areas, electrical equipment should be dust-tight (NEMA Type 5).

3.2 Power Generating and Utilization Equipment

3.2.1 General. In general, all electrical installations shall conform to NFPA 70, National Electrical Code. Specific details on the hazards of internal combustion engines, gas turbines, generators and transformers are covered in the NFPA, Fire Protection Handbook, and the FM, Loss Prevention Data Sheets.

3.2.2 Stationary Combustion Engines and Gas Turbines. NFPA 37, Stationary Combustion Engines and Gas Turbines, shall be followed for internal combustion engine and gas turbine and generator installations. Unless located in sprinklered areas the following shall apply:

a) Isolated installations may be protected with a self-contained pressurized sprinkler or water spray system with adequate supply to maintain a density of 0.2 gpm/ft^2 (8 l/min/m^2) for 2,000 ft^2 (184 m^2) for 40 minutes.

b) Where a turbine is installed as an individual unit in an enclosure, a carbon dioxide system is acceptable as primary protection in accordance with the following criteria for acceptance:

(1) For turbine compartments, the rate of discharge shall be such that a 34 percent concentration of carbon dioxide is attained within 1 minute after actuation and a 30 percent concentration shall be maintained throughout the deceleration period and until all metal surfaces likely to reignite oil are cooled to below the autoignition temperature of the oil 400-500° F (204-260° C).

(2) For accessory compartments, to extinguish fires in electrical equipment and oil, a concentration of 50 percent of carbon dioxide shall be obtained within 1 minute after actuation. No less than 30 percent concentration shall be maintained for the following 10 minutes.

(3) Where it can be established that fire temperatures will be below the decomposition temperature of Halon 1301 (under 900° F [482° C]), Halon 1301 shall be acceptable as an extinguishing system with provisions that the Halon discharge is sufficient to maintain a minimum design concentration of 5 percent throughout the deceleration period of the turbine.

c) Large individual units over 25,000 hp (18.63 MW) shall have backup protection of automatic sprinklers.

d) Emergency generators shall be protected by a local application of carbon dioxide.

3.3 Liquid-Insulated Transformers

3.3.1 Oil-Insulated Transformers. Fire protection equipment needed for outdoor transformers depends on their size and importance. Table 7 provides

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Table 7
Fire Protection for Outdoor Oil-Insulated Transformers

SIZE (EACH TRANSFORMER)	NUMBER	PROTECTION
Under 9,999 kVA	One or more	Portable Extinguishers
10,000 – 99,999 kVA [1]	One only	Hydrant Protection
10,000 – 99,999 kVA [1]	More than one	a. Provide a 25 – ft (8 m) minimum clear space between units, or b. Noncombustible barriers between units, or c. Fixed automatic waterspray
100,000 kVA and above	One or more	100,000 kVA and above. One or more fixed automatic water sprays Fixed automatic water spray

[1] Where there are mission – essential bus structures exposed to a possible transformer oil fire, or electric service or production could be interrupted for an extended period, a fixed automatic water spray system shall be provided to minimize the physical damage from fire and reduce the downtime for repairs.

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fire protection requirements for single-phase or polyphase outdoor transformers.

Buildings located 25 ft (8 m) or more from an oil-cooled transformer do not require fire exposure protection. Buildings within 25 ft (8 m) of a transformer require exposure protection by one of the following methods:

a) Protect transformers with a permanently piped water spray system.

b) Where transformers expose only one building of masonry or concrete construction, the following safeguards shall be provided where applicable:

(1) There shall be no window openings in first story walls within a horizontal distance of 10 ft (3 m) from the transformers. Existing window openings shall be closed using brick or concrete block.

(2) Window openings in the first story beyond 10 ft and up to 25 ft horizontally from the transformers shall be protected, using either wired-glass in steel sash or glass block.

(3) Window openings in second and third story walls directly above the transformers shall be protected using either wired-glass in steel sash or glass block.

(4) Overhanging eaves, where they exist, shall be fireproof.

c) Where transformers must expose a building having walls other than masonry or concrete, the following safeguards shall be taken:

(1) Provide a masonry or concrete barrier between building and transformers, with wing walls at each end of the barrier. The barrier shall extend at least 1 ft (0.3 m) above the top of transformer bushings and pressure-relief vents. The wing walls shall be of the same height and shall extend horizontally 2 to 3 ft (0.6 to 0.9 m) beyond the transformers, including any radiators and tap-changer enclosures. At multistory buildings, provide a fire-resistive roof on this three-sided transformer enclosure.

d) Where transformers expose more than one building, such as transformers located in courts or angles between buildings or where yard space is limited, they shall be enclosed in a suitable fire-resistive vault and protected as given in Table 8.

e) Where transformers are located on or above noncombustible roofs, suitable curbed and drained concrete mats or welded steel pans shall be underneath units and located so as not to expose roof structures. Oil-filled transformers shall not be installed on combustible roofs.

f) Oil-insulated transformers installed indoors shall be located in fire resistive vaults except for indoor transformer installations exempted by NFPA 70, National Electrical Code. Fire protection shall be provided in accordance with Table 8. Requirements apply to single-phase or polyphase lighting or power transformers.

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Table 8
Protection for Oil-Insulated Transformers in Fire-Resistive Vaults

SIZE (EACH TRANSFORMER)	NUMBER	FIRE PROTECTION
50,000 kVA and below	One or more, not exceeding 50,000 kVA total	Hydrant Protection
50,000 kVA and below	More than one, exceeding 50,000 kVA total	Automatic sprinkler, water – spray, or carbon dioxide system
Over 50,000 kVA	One or more	Automatic sprinkler, water – spray, or carbon dioxide system

3.3.2 Less Flammable, Liquid-Insulated Transformers. Transformers insulated with less flammable liquids (fire point of not less than 572° F (300° C), as defined by ASTM D-92-72) shall be permitted to be installed without a vault in Type I and Type II (see Table 1) buildings in areas in which no combustible materials are stored, there is a liquid confinement area, and the installation complies with all restrictions provided for in the listing of the liquid. Such indoor installations not meeting the requirements of the liquid listing, or installed in other than Type I or Type II buildings, or in areas where combustible materials are stored shall (1) be provided with an automatic fire extinguishing system and a liquid containment area, or (2) be installed in an approved vault. Transformers installed indoors and rated over 35,000 V shall be installed in a vault.

Transformers installed outdoors shall adhere to the same requirements for ordinary oil-insulated transformers. If they are isolated and present no exposure hazard to important structures, they may be protected with fire hydrants. For additional information, refer to the FM Data Sheets.

3.4 Trash Collection and Disposal Facilities

3.4.1 Requirements. Central trash collection units shall be placed 15 ft (4.6 m) or more away from wood frame or metal buildings or from openings in masonry-walled buildings. Collection, baling and storage room shall have 2-hr fire resistance and be sprinklered.

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Section 4: SPECIAL OCCUPANCIES AND HAZARDS

4.1 Purpose. The purpose of this section is to address certain specific occupancies and hazards to which attention must be drawn such that appropriate standards may be cited.

4.2 Personnel Support Facilities

4.2.1 Bachelor Quarters. This occupancy includes barracks, dormitories, sleeping quarters and similar facilities.

4.2.1.1 Requirements. This para. cross references para. 15-78 of AFR 88-15, Criteria and Standards for Air Force Construction. NFPA 101, Life Safety Code, provides requirements for life safety features. Installation of fire detection systems shall conform with the requirements of NFPA 101 and NFPA 72E except as modified in this handbook. Minimum requirements are:

a) A single station smoke detection unit shall be provided for each sleeping room of personnel housing quarters. Upon activation, an audible alarm in the room will be generated by the affected unit. Power to smoke detectors shall be hard-wired to the building electrical system. Battery powered detectors are not permitted.

b) Smoke detectors shall be located in interior corridors, lounges, and open bay sleeping areas of hotel-style personnel housing quarters. When activated, these units will sound a general building alarm and transmit a coded signal to a constantly monitored central alarm location. For Army and Air Force projects, a heat detector shall be provided in each sleeping room to sound the building fire alarm when activated.

c) Smoke detectors shall be located in living room/lounge areas of apartment-style personnel housing quarters. When activated, these units will sound a general building alarm and transmit a coded signal to a constantly monitored central alarm location.

d) Corridor detectors are not required if the building is provided with an automatic sprinkler system.

4.2.2 Family Housing. This section includes one-family, two-family, and multiple-family public quarters and rental housing.

4.2.2.1 Requirements. Fire protection criteria shall conform to the standards of the U.S. Department of Housing and Urban Development (HUD) 4900.1, Minimum Property Standards for One and Two Family Dwellings, and 4910.1, Minimum Property Standards for Multi-Family Housing, NFPA 101, Life Safety Code, and MIL-HDBK-1190, Facility Planning and Design Guide. Refer to NFPA 74, Household Warning Equipment, for additional guidance on fire detection systems.

4.2.3 Dining/Food Preparation Facilities. This para. cross references para. 15-78 of Air Force Regulation (AFR) 88-15, Criteria and Standards for Air Force Construction. Fire protection criteria shall conform to NFPA 96,

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Removal of Smoke and Grease-Laden Vapors from Commercial Cooking Equipment, and NFPA 101.

4.2.4 Administrative Facilities. Buildings and portions of buildings used primarily for offices are considered in this category. NFPA 101 contains requirements for this occupancy with respect to life safety features.

4.2.5 Hospitals and Dispensaries. This para. cross references para. 15-86 of Air Force Regulation (AFR) 88-15, Criteria and Standards for Air Force Construction. Hospitals and dispensaries shall comply with DOD requirements, the UBC, and the following NFPA standards:

a) NFPA 56A Inhalation Anesthetics
b) NFPA 56B Respiratory Therapy
c) NFPA 56C Laboratories in Health Related Institutions
d) NFPA 56D Hyperbaric Facilities
e) NFPA 56E Hypobaric Facilities
f) NFPA 56F Nonflammable Medical Gas Systems
g) NFPA 70 National Electrical Code
h) NFPA 76A Essential Electrical Systems for Health Care Facilities
i) NFPA 76C Safe Use of High Frequency Electricity in Health Care Facilities
j) NFPA 101 Life Safety Code

4.2.6 Detention and Correctional Facilities

4.2.6.1 Gang Release Devices. Mechanical or closed-circuit electrical gang release devices shall be provided whenever 12 or more locks must be operated to release prisoners confined in cells. This requirement does not apply to the dormitory prisoner population. All doors necessary to evacuate cell inmates shall be opened by the gang release device. Where gang release devices are not required by these criteria, all exit and corridor doors that must be opened for evacuation in case of fire shall be fitted with heavy, identically keyed, prison-type locks.

4.2.6.2 Hardware. Department of the Army, Corps of Engineers, Guide Specification CEGS-08701, Hardware: Prison-Locking Devices, shall apply.

4.2.6.3 Interior Finish. Interior finish including padded cells shall be Class A as defined by NFPA 101, Life Safety Code, with the exception that the smoke-developed rating shall not exceed 50, as determined by ASTM E-84, Standard Test Method for Surface Burning Characteristics of Building Materials.

4.2.7 Libraries. Combustible shelving, stacks, cases, cabinets, fixtures, furniture, and furnishings shall be eliminated to the maximum practicable extent. Wall and ceiling finish materials shall be noncombustible. Depending upon various circumstances associated with the library, the following special requirements shall be imposed:

a) All libraries located in sprinklered buildings shall be protected by sprinklers.

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b) In buildings not equipped with sprinkler protection, the following protection features shall be provided for those libraries containing substantial numbers of items that are rare, irreplaceable, important to activity mission:

(1) Buildings of fire resistive or noncombustible construction shall have smoke detection installation throughout the library area and provide a fire cutoff having a fire resistant rating of at least 1 hour to separate the library from other occupancies.

(2) Buildings of combustible construction shall have complete automatic sprinkler protection for the library area. The library area shall be separated by firewalls having a fire resistance rating of at least 2 hours.

(3) Where sprinkler protection is provided in accordance with item (2) above, wet-pipe systems with 212° F (100° C) heads shall be used. In special cases where storage arrangements and nature of library materials justify the additional cost, preaction systems may be used. NFPA 910, Libraries and Library Collections, and NFPA 911, Museums and Museum Collections, shall be applied where appropriate.

4.3 Electronic Equipment Installations This para. cross references para. 15-75 of AFR 88-15, Criteria and Standards for Air Force Construction.

4.3.1 Scope. The following minimum requirements are applicable to all major electronic equipment installations of high monetary value that are absolutely necessary to operations, weapons and communication systems, whether Government-owned or leased, including EDP, ADP, computers, avionics, communication, flight simulators, major training devices, etc. Electronic computer systems shall be any electronic digital or analog computer, along with all peripheral, support, memory, programming and all other directly associated equipment, records and storage. Word processing equipment, personnel computers, and Computer Aided Design (CAD) systems are not within the scope of this section.

4.3.2 Construction. New facilities to house electronic equipment installations shall be fire-resistive or noncombustible construction and separated from other occupancies by minimum 1-hr fire-rated construction. Additional guidance for construction and for protection is found in NFPA 75, Computer Data Processing Equipment.

4.3.2.1 Combustible Construction. Existing combustible construction shall be replaced with noncombustible construction and separated from other occupancies by fire-rated walls or partitions wherever practical.

4.3.3 Sprinkler Systems. Standard wet-pipe or preaction sprinkler systems with 212° F (100° C) heads shall be provided in the following areas:

a) Electronic equipment areas where combustible materials of construction are present or where combustible materials, including cards, paper, and plastics, are processed or stored (excluding cards, paper, disks, and tapes within machines).

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b) Areas containing flight simulators or major electronic training devices where combustible materials are present in the basic structure or as part of the simulator trainer.

c) Record storage rooms unless records are contained in closed metal files or cabinets.

d) Areas adjacent to electronic equipment spaces in building of noncombustible or fire-resistive construction where the occupancy in such areas presents fire exposures to the electronic equipment spaces.

4.3.4 Gaseous Extinguishing Systems

4.3.4.1 National-Defense Equipment Spaces. Halon 1301 total flooding systems may be provided in electronic equipment spaces where strategically important functions vital to our national defense are conducted, and continued operation of the equipment is absolutely necessary to ongoing strategic military operations (e.g., command control centers, weapon control systems and communication facilities, but excluding administrative applications such as word processors, personnel computers, personnel records, inventory records and payroll records). Installation of total flooding Halon 1301 systems shall not be considered as a substitute for required sprinkler systems.

NOTE: For strategically important functions it can only be assumed that backup capability is provided at other locations, thus reducing the dependence upon the primary equipment. In such cases, and assuming the equipment is manned 24 hours a day for 365 days a year, adequate fire extinguishers would be an alternative to a total flooding gaseous fire extinguishing system.

4.3.4.2 Underfloor Spaces. Halon 1301, carbon dioxide or automatic sprinkler systems shall be provided for the underfloor spaces unless the following conditions apply:

a) Power cables are either type MI, MC or AC cable or it is installed in approved conduit or metallic tubing.

b) Communication cables, data cables and ADP interconnecting cables are installed in approved conduit or metallic tubing or listed as approved for use in air plenums.

4.3.4.3 High-Cost Electronic Training Devices. Halon 1301 or carbon dioxide shall be provided to protect against hydraulic fluid fires in flight simulators or other high cost electronic training devices when Class A combustibles are not present and automatic sprinkler protection is not required.

4.3.5 Automatic Fire Detection Systems

4.3.5.1 Electronic Equipment and Record-Storing Spaces. Smoke detection systems shall be provided in electronic equipment areas and in record storage rooms. Detectors shall transmit a signal to the fire department or central alarm location.

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4.3.5.2 Underfloor Cable Spaces. Smoke detection systems shall be provided in underfloor spaces containing exposed nonmetallic sheathed cables. Fixed-temperature rate- compensated or fixed temperature rate-of-rise heat detectors may be provided in underfloor spaces under circumstances when smoke detectors are not effective. Systems shall transmit signal to fire department or central alarm location and shall activate underfloor Halon 1301 or carbon dioxide systems. Alarms may be operated by the cross zoned concept. Automatic operation of the underfloor gaseous fire extinguishing system may be deleted; manual activation is permitted if the space is continously manned and/or the equipment is not considered vital to the national defense.

4.3.6 Emergency Power Shutoff Switches. Where required, such switches shall be located at the principal exits.

4.4 Industrial Facilities

4.4.1 Overhaul and Repair Shops. Facilities falling within this category are those in which major overhaul and repairs are made to various types of equipment or their component parts. Disassembly and testing may also be performed in such facilities, which include aircraft shops, automotive garages and repair shops, ship repair shops, and ordnance shops.

4.4.1.1 Requirements. Automotive garages shall conform to NFPA 88B, Repair Garages. The requirements for ship repair facilities in NFPA 303, Marinas and Boatyards, and NFPA 312, Fire Protection of Vessels During Construction, Repair, and Lay-up, shall be followed. Major aircraft overhaul and repair shops shall conform to NFPA 409, Aircraft Hangars, and to paras. 4.4.6 and 4.4.7 of this handbook.

4.4.1.2 Other Standards. Because repair shops may contain multiple occupancies, each including a specific hazard, the following additional standards shall be complied with where appropriate.

a) NFPA 33 Spray Application Using Flammable and Combustible Materials
b) NFPA 51 Oxygen-Fuel Gas Systems For Welding and Cutting.
c) NFPA 70 National Electrical Code

For other common and special hazards refer to the appropriate paragraph of this section.

4.4.2 Ordnance Production Facilities. Automatic sprinkler systems in ordnance facilities shall be provided with flexible couplings and sway bracing similar to that provided for buildings in earthquake zones. Specific automatic sprinkler protection requirements for areas within these structures are as follows:

a) High-speed (operation of 0.5 seconds or less), preprimed, deluge systems wherever exposed powder, explosives, or propellants are processed or stored. Complete protection of such locations is essential (refer to Table 10, Water Demands for Sprinklered Facilities in Section 5).

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b) Ordinary deluge systems in other areas or auxiliary sections of buildings in which processing or storage of explosives or propellants takes place.

c) Wet-pipe sprinkler systems in other areas or auxiliary sections of buildings if separated by fire partitions.

d) Wet-pipe or preaction sprinkler systems where missile assembly inspection or storage is carried on and where the propellant is confined within the missile or warheads are present. Similar protection shall be provided for torpedo and air underwater weapons shops. If electronic equipment is present in assembly and inspection areas, preaction sprinkler systems may be provided.

e) Heat detection equipment of any type is acceptable if it meets the operating time limitations and is suitable in other respects, such as complying with explosion-proof requirements. When pneumatic-type detection equipment is used, not more than three detectors, and preferably only one, shall be on a single circuit. They must be in the same heat influence area. Photoelectric detectors may be used for spot or for local protection applications, but shall not substitute for area or room heat detectors.

f) A system providing complete supervision of all deluge and preaction systems shall be provided so that any deficiency that develops that would affect the speed or reliability of operation will give a distinctive alarm separate from the water flow alarm.

g) Water supplies for deluge systems shall be adequate to supply the total demand of the largest fire area at the specific residual pressure required by the system, plus an allowance for hose stream demand, for a period of 45 minutes.

4.4.3 Storage. Consideration must be given to proper storage of materials to prevent the undue hazard and concentration of large quantities in a single location subject to a single catastrophe.

4.4.3.1 Outside Storage. Material stored outside shall comply with the following NFPA standards:

a) NFPA 30 Flammable and Combustible Liquids Code
b) NFPA 46 Storage of Forest Products

Water supply requirements shall be in accordance with para. 5.2.2.5 of this handbook.

4.4.3.2 Inside Storage. Inside storage arrangements and facilities shall comply with the following NFPA standards:

a) NFPA 30 Flammable and Combustible Liquids Code
b) NFPA 231 Indoor General Storage
c) NFPA 231C Rack Storage of Materials
d) NFPA 231D Storage of Rubber Tires
e) NFPA 232 Protection of Records

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4.4.4 Waterfront Facilities. Waterfront and harbor facilities shall comply with the following:

a) NAVFAC DM-25 Series on Piers and Dockside Facilities
b) NAVFAC DM-29 Series on Drydocks and Marine Railways
c) NFPA 307 Marine Terminals, Piers and Wharves
d) NFPA 312 Fire Protection of Vessels During Construction, Repair and Lay-up

4.4.5 Petroleum Oils and Lubricants (POL) Facilities. Requirements for POL facilities are contained in NFPA 30, Flammable and Combustible Liquids Code, and NFPA 385, Tank Vehicles for Flammable and Combustible Liquids, and applicable service criteria. The standards of the American Petroleum Institute (API) provide additional guidance. Refer to Section 5 of this handbook for water supply requirements. For Navy requirements, refer to NAVFAC DM-22, Petroleum Fuel Facilities, for additional criteria. For Air Force projects see NAVFAC DM-22 and AFR 88-15, Criteria and Standards for Air Force Construction for additional information.

4.4.5.1 Combustible Liquids Under Pressure (For Navy Projects Only). Any combustible liquid under pressure shall be treated as a flammable liquid when such liquids are located in pump houses and hydraulic test facilities. For pump houses refer to NAVFAC DM-22. For Hydraulic Test Systems, comply with the following:

a) For hydraulic systems that use pressures exceeding 200 psi (1379.5 kPa), SAE 1,010 dead-soft, cold-drawn, seamless-steel tubing (or equivalent) shall be used. A factor of safety of eight over normal working pressures shall be used. Tubing is preferable to pipe. It can be bent to fit in restricted spaces with a minimum number of fittings, reducing the number of possible leakage points. Solderless, steel fittings of the flareless "locking-sleeve" type or flare type shall be used.

b) The use of threaded pipe should be avoided. Where threaded connections are used, the requirements of ANSI B2.1 shall be met. A safety factor of eight over maximum normal working pressures shall be used.

c) Tubing runs shall have as few bends as possible, but should have at least one bend to provide for thermal expansion and contraction. The minimum radius of tube bend shall be three tube diameters.

d) Where hose must be used for flexible connections, it shall be steel reinforced, designed for the hydraulic fluid being used, and capable of withstanding five times the actual operating pressure. Hose couplings and fittings and minimum bending radius shall be in accordance with the hose manufacturer's instructions. Hose shall be installed so as not to rub against objects as a result of machine movement, vibration or pressure surges.

e) Piping and tubing shall be anchored or secured to

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minimize failure due to vibration. Pipe supports shall not prevent normal thermal expansion.

f) There shall be an accessible, well-marked, emergency shut-off switch for each pump.

g) Provide automatic shut off switch to deactivate hydraulic pump upon loss of pressure.

4.4.5.2 Petroleum-Based Hydraulic Fluids. The following shall apply, in addition to the above requirements, to petroleum-base hydraulic oils in nonmobile equipment:

a) Provide automatic sprinklers over and at least 20 ft (6.1 m) beyond the hydraulic equipment. Complete sprinkler is required if the structure is of combustible construction. Sprinklers may be omitted over a single small system or multiple adjacent small systems not exceeding 100 gal (378.5 l) aggregate capacity and if the construction is non-combustible and ignition sources are not normally present and provisions exist for automatic or manual shut down of the system.

b) An automatic switch, activated by sprinkler waterflow alarm, fusible link or other fire detector, shall be provided to shut down the system if there is 100 gal or more of oil.

c) All electrical equipment installed within 25 ft (7.62 m) of hydraulic pumps, meters, fittings, meters and other similar devices located in the system's lines shall be Class 1, Division 2. All electrical equipment and devices within this area (including transformers, capacitors, meters, relays, instruments, switches, light fixtures and motor control equipment) shall be UL listed as suitable for installation in Class 1, Division 2 environments as defined in the latest edition of the National Electrical Code, Article 501. Wiring within these areas shall be in threaded, rigid, metal conduit, with seals installed to separate hazardous and non-hazardous areas.

4.4.6 Aircraft Hangars. This para. cross references para. 15-74 of AFR 88-15, Criteria and Standards for Air Force Construction. Requirements listed in this section are applicable to fuel cell maintenance facilities, corrosion control and protective coating, and general purpose maintenance hangars. All hangars shall comply with NFPA 409, Aircraft Hangars, except where modified as follows:

a) Aqueous Film-Forming Foam (AFFF) deluge sprinkler systems or preaction AFFF sprinkler systems using standard water sprinkler heads shall be provided in main hangar areas. The discharge rate shall be a minimum of 0.16 gal (0.61 l) of air-foam solution per minute per ft^2 (6 $l/min/m^2$) floor area. For Navy projects, preaction sprinkler systems shall only be used when supplementary low level fixed or oscillating foam nozzles are provided.

b) Where preaction AFFF sprinkler systems are used, water supply in addition to hose streams and monitor nozzles shall be sufficient to satisfy the volume of water required by the number of sprinkler heads opened in accordance with one of the following:

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(1) Within an 8,000 ft^2 (740 m^2) area where roof or ceiling height is 25 ft (7.6 m) or less above the floor level.

(2) Within an 18,000 ft (1670 m^2) area where roof or ceiling height is over 25 ft but not more than 75 ft (23 m) above the floor level.

(3) Within a 35,000 ft (3250 m^2) area where roof or ceiling height is over 75 ft above the floor level.

c) Supplementary low-level, fixed, AFFF nozzle systems shall be provided in main hangar areas. Oscillating nozzles may be provided instead of fixed nozzles if the use of fixed nozzles is not feasible. These systems shall be designed at a minimum application rate of 0.10 gal (0.38 l) per minute per ft^2 (4 l/min/m^2). These systems shall be activated by the overhead detection system. Strategically located manual controls shall also be provided.

NOTE: Supplementary low-level systems are not required in hangars that are strictly dedicated to helicopters and small aircraft if hose lines are provided, if helicopters are not strategically important, and if the shadow area under the helicopter is not excessive.

d) Foam concentrate tanks, proportioning equipment, and deluge valves shall be separated from main hangar areas by construction having a minimum of 1-hr fire resistance rating.

e) Automatic, wet-pipe sprinkler systems shall be provided in all areas of the hangar facility not requiring overhead AFFF sprinkler protection.

4.4.7 Aircraft Acoustical Enclosures

4.4.7.1 Complete Enclosures (Hush-House). This para. cross references para. 15-95 of AFR 88-15, Criteria and Standards for Air Force Construction. Requirements shall be the same as those listed for Aircraft Hangars, para. 4.4.6, in addition to the following:

a) Overhead AFFF systems shall be designed for the entire floor area.

b) Low-level systems shall provide uniform coverage for the entire floor area.

c) The overhead and the low level systems shall both be automatically actuated by fixed temperature-rate compensation detectors. Switches for interrupting the automatic features shall be provided in the control room. Clearly visual indication shall be provided when the automatic feature has been interrupted.

d) Separate manual controls for actuation of each aqueous film-forming foam system shall be provided in the control room.

4.4.7.2 Out-of-Airframe Acoustical Enclosures (Test Cells). The requirements for complete acoustical enclosures listed in para. 4.4.7.1, shall be provided except for the following:

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a) In lieu of the overhead AFFF system required in paras. 4 through 4.7.1, overhead water deluge protection having a discharge density of 0.35 gal (1.3 l) per minute per ft^2 (14 l/min/m^2) of the entire floor area may be used where adequate water supply is available and where this type of extinguishing system is effective in controlling fuel that may be encountered.

b) In lieu of the supplementary low level AFFF system described in para. 4.4.6, a supplementary local application system for engine protection shall be provided. Engines shall be protected by local application of water spray with a density of 0.50 gal (1.89 l) per minute per ft^2 (20 l/min/m^2) of engine surface area. Where conditions warrant, local application of carbon dioxide or Halon 1301 fire extinguishing systems may be used as an alternative to water spray systems.

c) The fuel room shall be protected by deluge or wet-pipe sprinkler system.

4.4.8 Hyperbaric and Hypobaric Chambers. Hyperbaric chambers shall conform to NFPA 99, Chapter 10. Hypobaric chambers shall conform to NFPA 99, Chapter 11.

4.4.9 Anechoic Chambers. Anechoic chambers shall be protected with automatic sprinkler systems in accordance with Table 9. When chamber construction materials used to eliminate echoes and reverberations have passed the FM Research Corporation Corner Test, sprinklers are not required.

4.5 High Rise Buildings. This para. cross references para. 15-84 of AFR 88-15, Criteria and Standards for Air Force Construction. Buildings in which the upper floors are beyond the reach of fire department ladders or aerial equipment (normally 75 ft [23 m]) and in which fires must be fought from the inside and present serious life loss potentials from fire and severe difficulties in firefighting operations. Such buildings require the following:

a) Automatic sprinkler protection in accordance with Chapter 8 of NFPA 13, Installation of Sprinkler Systems.

b) Smoke control systems shall be provided in buildings where the windows cannot be opened.

c) Means for placing control of elevators under emergency forces complying with the requirements of ANSI A17.1B, Safety Code for Elevators, Dumbwaiters, Escalators and Moving Walks.

d) Building control center with two-way communications to all floors, elevators, and stair towers.

4.6 Flammable and Combustible Liquids

4.6.1 Specific Safeguards. The following standards and guides are applicable to the storage, handling and use of flammable and combustible liquids and shall be applied:

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Table 9
Fire Protection for Anechoic Chambers[1]

PROTECTION	SMOOTH SURFACES		GEOMETRIC SHAPES (PYRAMIDS)		
	UP TO 10 In. THICK (250 mm)	OVER 10 In. THICK (250 mm)	UP TO 6 In. THICK (150 mm)	6 TO 12 In. THICK (300 mm)	OVER 12 In. THICK (300 mm)
Automatic sprinklers (100) ft^2 coverage, 10 – ft max spacing, 212 ° F, 1/2 in. extra hazard piping, 0.30 gal/min/ft^2 (12 l/min/m^2) average density [2]	X		X		
Automatic sprinklers (50 ft^2 coverage, 8 – ft max spacing, 212 ° F, 1/2 in. extra hazard piping, 0.60 gal/min/ft^2 (25 l/min/m^2) average density)		X		X[3]	
Automatic sprinklers (40 ft^2 coverage, 7 – ft max spacing, 212° F 1/2 in. extra hazard piping, 0.60 gal/min/ft^2 (25 l/min/m^2) average density					X[3]
Side wall automatic sprinklers (not over 7 – ft spacing)	X[4]	X[4]	X[3,5]		
Side wall automatic sprinklers (not over 5 – ft spacing)				X[3,5]	X[3,5]

1 Adapted from FM, Loss Prevention Data Sheets.

2 Average Density – At chambers under 2,000 ft^2 (186 m^2) of floor area, the average density shall be for all heads. At chambers over 2,000 ft^2, average density shall be for 3/4 of the sprinklers.

3 Sprinklers shall be located 1/3 the length of the pyramid from the base of the pyramid.

4 Required for walls over 15 ft (5 m) high and installed at approximately 15 ft vertical intervals.

5 Intermediate row required for walls over 15 ft high.

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a) NFPA 30 Flammable and Combustible Liquids Code
b) NFPA 33 Spray Application Using Flammable and Combustible Materials
c) NFPA 34 Dipping and Coating Processes Using Flammable or Combustible Liquids
d) NFPA 68 Explosion Venting Guide
e) NFPA 407 Aircraft Fuel Servicing
f) NFPA 415 Aircraft Fueling Ramp Drainage

4.7 Flammable Gases

4.7.1 Specific Safeguards. The following standards and guides are applicable to the storage, handling, and use of hazardous gases and shall be applied:

a) NFPA 50 Bulk Oxygen Systems at Consumer Sites
b) NFPA 50A Gaseous Hydrogen Systems at Consumer Sites
c) NFPA 50B Liquefied Hydrogen Systems at Consumer Sites
d) NFPA 51 Oxygen-Fuel Gas Systems for Welding and Cutting
e) NFPA 54 National Fuel Gas Code
f) NFPA 56A Inhalation Anesthetics
g) NFPA 56B Respiratory Therapy
h) NFPA 56D Hyperbaric Facilities
i) NFPA 56E Hypobaric Facilities
j) NFPA 56F Nonflammable Medical Gas Systems
k) NFPA 58 Liquefied Petroleum Gases
l) NFPA 59A Liquefied Natural Gas

4.8 Hazardous Chemicals

4.8.1 Specific Safeguards. The following standards are applicable to the storage, handling or use of specific hazardous chemicals and shall be applied:

a) NFPA 43A Storage of Liquid and Solid Oxidizing Materials
b) NFPA 48 Storage, Handling and Processing of Magnesium
c) NFPA 481 Production, Processing, Handling and Storage of Titanium
d) NFPA 482M Production, Processing, Handling and Storage of Zirconium
e) NFPA 490 Storage of Ammonium Nitrate
f) NFPA 492 Separation Distances of Ammonium Nitrate and Blasting Agents
g) NFPA Hazardous Materials Handbook

Many hazardous chemicals are not covered by specific standards, but NFPA 49, Hazardous Chemical Data, includes suggestions and remarks pertinent to firefighting and storage for each of the chemicals listed. Whenever a specific standard is lacking, the recommendations contained in NFPA 49 shall be followed.

4.9 Explosives and Blasting Agents

4.9.1 Safeguards. Safeguards for the storage, handling and use of explosives and blasting agents other than military explosives shall be

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provided as required by NFPA 495, Explosive Materials, and for Navy projects additional criteria are found in NAVSEA OP-5, Ammunition and Explosives Ashore.

4.10 Combustible Dusts

4.10.1 Specific Safeguards. The following standards and guides are applicable to the storage, handling, use, or production of combustible dusts and shall be applied:

a) NFPA 65 Aluminum, Processing and Finishing
b) NFPA 68 Explosion Venting
c) NFPA 69 Explosion Prevention Systems
d) NFPA 85F Pulverized Fuel Systems
e) NFPA 91 Blower and Exhaust Systems
f) NFPA 651 Manufacture of Aluminum or Magnesium Powder
g) NFPA 654 Prevention of Dust Explosion in Plastics Industry
h) NFPA 664 Prevention of Dust Explosions in Woodworking and Wood Floor Manufacturing Plants

4.11 Ovens and Furnaces

4.11.1 Safeguards. Ovens and furnaces shall be constructed, installed, and protected in accordance with the following NFPA standards:

a) NFPA 86A Ovens and Furnaces
b) NFPA 86B Industrial Furnaces
c) NFPA 86C Industrial Furnaces Special Processing

4.12 Hazardous Materials

4.12.1 General Discussion. The storage of hazardous materials and the storage of hazardous wastes are two distinct and separate types of facilities and each has its own set of criteria. For criteria governing hazardous storage refer to MIL-HDBK 1032/2, Covered Storage, and for hazardous waste refer to MIL-HDBK-1005/13, Hazardous Waste Storage Facilities.

4.12.2 Hazardous Waste Storage Facilities. There are two types of hazardous waste storage facilities: short-term storage facilities for waste stored for less than 90 days, and long-term storage facilities for materials stored for more than 90 days. The short-term facility does not require a permit to operate but must conform to the requirements given in 40 CFR 262.34. The long-term facility is a permanent facility subject to the requirements of CFR 40 Series, Protection of Environment, CFR 40-264, Regulations for Owners and Operators of Permitted Hazardous Waste Facilities, and CFR 40-265, Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities, and the permit requirements of CFR 40-270, EPA Administered Permit Program: The Hazardous Waste Permit Program. Other applicable criteria for the design of hazardous waste storage facilities are as follows:

a) DOD 4145.19-R-1, Storage and Handling of Hazardous Materials

b) DOD Instruction 6050.5, Hazardous Materials Information System (HMIS)

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c) PL-580, Subtitle C, as regulated by 40 CFR, Part 124, 260 through 266 and 270, Resource Conversation and Recovery Act

d) PL 96-510, The Comprehensive Environmental Response Compensation and Liability Act of 1980

e) 29 CFR 1910, OSHA Safety and Health Standards

f) National Fire Protection Association (NFPA):

(1) NFPA 30 Flammable and Combustible Liquids
(2) NFPA 43C Storage of Gaseous Oxidizing Materials
(3) NFPA 43D Storage of Pesticides in Portable Containers
(4) NFPA 49 Hazardous Chemicals Data
(5) NFPA 490 Storage of Ammonium Nitrate

g) 49 CFR (Parts 171 through 179), Department of Transportation Requirements

h) (40 CFR 300) National Oil and Hazardous Substances Pollution Contingency Plan

i) (40 CFR 761) Toxic Substance Control Act Regulations

4.12.2.1 Fire Protection for Hazardous Waste Storage Facilities. The following minimum criteria shall be provided:

a) Exterior Firewalls:

(1) Exterior exposing firewalls shall be a 4-hour fire rating when the facility is attached to a structure or it is located within 10 ft (3 m) or less from another building or property line.

(2) Exterior exposing firewalls shall be of 2-hour fire rating when the facility is located more than 10 ft but less than 50 ft (15m) from an important building or property line.

(3) The exterior walls shall be of noncombustible construction when the exposed buildings or property line is over 50 ft from the facility.

b) Interior Firewalls:

(1) Interior firewalls shall be of 4-hour construction if the facility is located within a structure that houses other occupancies.

(2) Interior firewalls shall be of 2-hour construction when the area of the room exceeds 300 ft^2 (27 m^2).

(3) Interior firewalls shall be of 1-hour construction when the area of the room is 300 ft^2 or less.

c) Containment. Water discharge from the sprinkler system need not be contained except for that volume that would be trapped by the containment

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systems required in MIL-HDBK 1005/13, Hazardous Waste Storage Facilities.

d) Sprinkler Protection. The sprinkler systems installed shall be suitable for a corrosive environment. The storage areas for reactive wastes shall be protected by an automatic operating fixed gaseous type fire protection system. The floor densities for the various storage areas shall be 0.35 gpm/ft^2 (14.2 lpm/m^2) in the waste storage areas and 0.16 gpm/ft^2 (6.56 lpm/m^2) in the support areas. Exception: Refer to NFPA 30 for densities required in flammable and combustible liquid storage areas.

NOTE: The use of AFFF shall be investigated when the water supplies may be marginal or where Class I flammable liquids are stored.

e) Explosion Venting. Explosion venting shall be provided in the organic, reactive, and general waste storage areas.

f) Explosion-Proof Electrical Equipment. Such equipment shall be provided in the organic, reactive, oxidizer and general waste storage areas.

4.12.3 Hazardous Material Storage Areas. Fire protection for these areas shall be the same as hazardous waste storage areas. For additional information refer to MIL-HDBK-1032/2, Covered Storage.

4.13 Additional/Supplemental Criteria for Air Force. AFR 88-15 contains additional/supplemental criteria for the following facilities:

a) Liquid oxygen and Anhydrous Ammonia Tanks.
b) Dependent Educational Facilities.
c) Hardened Aircraft Shelter and Servicing Areas.
d) Warehouses.
e) Aircraft Engine Test Facilities (Test Cells).
f) Child Care Centers.
g) Windowless Facilities or Underground Areas.
h) Vehicle Operations, Parking, Storage, Maintenance and Repair Facilities.

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Section 5: WATER SUPPLY FOR FIRE PROTECTION

5.1 Water Demands for Sprinklered Facilities

5.1.1 Factors Influencing the Water Demand for Sprinklers. The water demand required for sprinkler protection depends upon the occupancy (refer to Appendix B, Occupancy Hazard Classification), the discharge density, design area, and the type of sprinkler system (wet or dry), construction and other building features.

5.1.2 Water Demand for Sprinklers. The water demand required for sprinklers shall be determined from Tables 10 and 11. Refer to Appendix B for occupancy classifications.

5.1.2.1 Design Densities. The design densities indicated in Table 10 are minimum densities, and each sprinkler in the design area shall discharge at least the flow rate required to produce the stipulated density.

5.1.2.2 Design Area. The design areas shown in Table 10 are the hydraulically most remote areas.

5.1.3 Water Demand for Hose Streams. Hose streams are needed concurrently with sprinkler discharge in order to effect final extinguishment or to wet down adjacent structures. The hose stream demand for sprinklered occupancies shall be determined from Tables 10 and 11.

5.1.4 Total Water Demand for Sprinklered Occupancies. The total water demand for sprinklered occupancies is equal to the sum of the sprinkler system water demand and the hose stream demand. The total demand shall be available at the sprinkler system connection to the underground main, and at the pressure necessary to produce the required sprinkler density over the required area of sprinkler operation.

5.1.5 Water Demand for Sprinklers (Special Facilities). Special requirements apply to some facilities, as indicated in paras. 5.1.5.1 through 5.1.5.5.

5.1.5.1 Aircraft Hangars. Water demands for aircraft hangars shall be in accordance with Section 4 of this handbook.

5.1.5.2 Warehouses (High Piled or High Rack Storage). Water demands for warehouses containing high rack storage shall conform to NFPA 231C, Rack Storage of Materials. Water demands for warehouses containing high piled storage shall conform to NFPA 231, Indoor General Storage.

5.1.5.3 Ordnance Facilities. Refer to paras. 4.4.2 and 4.9.

5.1.5.4 Rubber Tire Storage. These occupancies shall be in accordance with NFPA 231E, Storage of Rubber Tires.

5.1.5.5 Aircraft Acoustical Enclosures. Refer to para. 4.4.7.

5.2 Water Demands for Unsprinklered Facilities

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Table 10
Water Demands for Sprinklered Facilities

OCCUPANCY CLASSIFICATION[1]	SPRINKLER		HOSE Gal/Min (l/Min)	DURATION OF SUPPLY (Minutes)
	DESIGN DENSITY[2] (Gal/Min)/Ft^2 (l/Min/m^2)	DESIGN AREA Ft^2 (m^2)		
Light Hazard	0.10 (4)	3,000 (280)	250 (960)	45
Ordinary Hazard Group 1	0.15 (6)	3,000 (280)	500 (1,920)	60
Ordinary Hazard Group 2	0.20 (8)	3,000 (280)	500 (1,920)	75
Ordinary Hazard Group 3	0.25 (10)	3,000 (280)	500 (1,920)	90
Extra Hazard	0.35 (14)	3,000 (280)	750 (2,820)	105
Ordnance Plants (exposed powder area)	0.50 (20)	Entire area	750 (2,820)	45
Rubber Tire Storage	0.60 (25)	3,000 (280)	750 (2,820)	150
Missile Assembly	0.35 (14)	3,000 (280)	750 (2,820)	45

1 Refer to Appendix B for occupancy definitions.

2 Use pipe schedule for the design of systems for areas less than 3,000 ft^2. Use pipe schedule or hydraulic calculations for uniform distribution of systems for areas larger than 3,000 ft^2. Use hydraulic calculations for open areas larger than 10,000 ft^2.

3 For dry pipe systems increase design area by 30 percent.

Table 11
Water Demands for Unsprinklered Facilities[1]

OCCUPANCY CLASSIFICATION (REFER TO APPENDIX B)	HOSE STREAMS[1] Gal/Min (l/Min) AT 20 lb/in.^{2} (140 KPa) RESIDUAL PRESSURE	DURATION[1] (Minutes)
Light Hazard	750 – 1,500 (2,820 – 5,700)	60 – 120
Ordinary Hazard Group 1	1,000 – 2,000 (3,780 – 7,560)	90 – 150
Ordinary Hazard Group 2	1,500 – 3,000 (5,700 – 11,340)	90 – 150
Ordinary Hazard Group 3	2,000 – 4,000 (7,560 – 15,120)	120 – 180
Extra Hazard	2,500 – 5,000 (9,480 – 18,900)	150 – 240

1 Ranges based upon favorability of firefighting conditions as defined in para 5.2 of this handbook.

Note: Refer to Appendix C for Army and Air Force Projects

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5.2.1 Hose Stream Demand for Unsprinklered Facilities. Water demands and duration for unsprinklered facilities shall be determined by Table 11 for Navy Projects and Appendix C for Army and Air Force Projects.

The ranges of water demands and duration for unsprinklered occupancies shall be determined by Table 11. The following factors affect these ranges and shall be used to determine the water flow demand:

a) Occupancy of classification,
b) Response time by fire department,
c) Type of construction,
d) Number of stories,
e) Separation distances,
f) Building floor area, and
g) Firefighting access.

NOTE: In general, when the required fire flow rate exceeds 2,500 gpm (9,462 lpm), a benefit/cost analysis shall be conducted to determine if additional fire protection systems or if a change in the construction type or use of a facility is more cost effective than providing the required flow. Such changes may reduce the required flow to a point where it can be provided more economically.

5.2.2 Hose Stream Demand for Special Facilities. Special requirements may apply to certain facilities. Such facilities include ship berthing and drydock facilities, dwellings, POL areas, aircraft parking/refueling areas, vehicle parking areas and yard storage.

5.2.2.1 Ship Berthing and Drydock Facilities. The water demand for ship berthing and drydock facilities shall be as required by NAVFAC DM-25 Series, Piers and Dockside Facilities, and DM-29 Series, Drydocks and Marine Railways.

5.2.2.2 Dwellings. The water demand for dwellings shall be as follows:

a) One-story - 500 gal. per minute (1,920 l/min) for 90 minutes.
b) Two-story - 750 gal. per minute (2,820 l/min) for 90 minutes.
c) Three-story and above - 1,000 gal per minute (3,780 l/min) for 90 minutes.

5.2.2.3 Petroleum Oils and Lubricants (POL) Areas. POL areas shall conform to the following:

a) Above Ground Atmospheric POL Tanks. Table 12 provides fire flow rates for nonpressurized POL tanks.

b) Above Ground Pressurized POL Tanks. Table 13 provides fire flow rates for pressurized POL tanks.

5.2.2.4 Aircraft Parking and Refueling Facilities. A minimum fire flow rate of 1,000 gpm (3,780 l/min) for a 2-hr duration is to be provided for all such facilities.

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Table 12
Atmospheric POL Tank Cooling Water

TANK DIAMETER Ft	(m)	FIRE FLOW RATE gpm	(l/Min)
0 – 65	(0 – 20)	500	(1,920)
65 – 120	(20 – 36)	750	(2,820)
120 – 155	(36 – 47)	1,000	(3,780)
155 – 200	(47 – 61)	1,250	(4,740)
200 or greater	(61 –)	1,500	(5,700)
Minimum duration: 240 minutes.			

Note 1 Allow an additional 500 gpm (1,920 l/min) for each exposed tank, pressure vessel or handling facility within 50 ft (15 m) or one tank diameter, whichever is greater, of the largest tank under consideration.

Note 2 For Navy Projects the maximum water supply for storage tanks shall not exceed 2500 gpm.

5.2.2.5 Yard Storage. Water demand for yard storage is presented in Table 14. Factors influencing water demand requirements for outside yard storage are:

a) Size of storage area - Three ranges of storage area are presented: 0-10,000 ft^2 (0-929 m^2); 10,000-25,000 ft^2 (929-2,322 m^2), and greater than 25,000 ft^2 (2,322 m^2).

b) Material combustibility hazard classification - Materials are classified according to the criteria used in NFPA Standard 231A, Recommended Safe Practices for Outdoor General Storage, which categorizes materials as noncombustible or combustible. Because of the increased fire hazard associated with high-piled storage, the combustible classification has been subdivided to reflect this.

c) Separation - Consideration has been made to include the effects of exposure to adjacent storage, buildings or structures. Separation factors are computed using the following methodology. First, the shortest distance between each exposed facility and the yard storage perimeter is determined.

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Table 13
Pressurized POL Tank Cooling Water

TANK GROUP SIZE	FIRE FLOW RATE gpm (l/Min)
Single tank (less than 30,000 gal capacity) (113 550 l)	250 (960)
Single tank (more than 30,000 gal capacity) (113 550 l)	500 (1,920)
2 to 6 tanks (one or more tanks greater than 30,000 gal capacity) (113 500 l)	500 (1,920)
2 to 6 tanks (each tank greater than 30,000 gal capacity) (113 500 l)	1,000 (3,780)
7 or more tanks, each tank less than 30,000 gal capacity (113 550 l)	1,000 (3,780)
7 or more tanks, one or more tanks greater than 30,000 gal capacity (113 500 l)	1,500 (5,700)

Minimum Duration: 240 minutes

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Table 14
Water Demand for Various Conditions of Yard Storage[1]

AREA/MATERIAL CLASSIFICATION	ISOLATED[2]		NON-ISOLATED[3]	
	FLOW (gpm) (l/min)	DURATION	FLOW (gpm) (l/min)	DURATION
I. 0 – 10,000 ft^2 (0 – 929 m^2)				
A. Noncombustible	500 (1,920)	120 min.	875 (3,312)	120 min.
B. Combustible, Low Piled	1,000 (3,780)	120 min.	1,750 (6,624)	180 min.
C. Combustible, High Piled	1,500 (5,700)	120 min.	2,625 (9,937)	180 min.
II. 10,000 – 25,000 ft^2 (929 – 2,322 m^2)				
A. Non combustible	750 (2,820)	120 min.	1,310 (4,960)	120 min.
B. Combustible, Low Piled	1,500 (5,700)	120 min.	2,625 (9,937)	180 min.
C. Combustible, High Piled	2,000 (7,560)	180 min.	3,500 (13,249)	180 min.
III. Greater than 25,000 ft^2 (2,322 m^2)				
A. Noncombustible	1,000 (3,780)	120 min.	1,750 (6,624)	180 min.
B. Combustible, Low Piled	2,000 (7,560)	180 min.	3,500 (13,249)	180 min.
C. Combustible, High Piled	2,000 (7,560)	180 min.	3,500 (13,249)	180 min.

1 The water demands of this table represent the water requirements of the upper limit of storage area of each category. For areas less than the maximum in each classification, refer to Appendix C, Figures C – 5 and C – 6 for water requirements. For water flow less than 1,750 gpm, use 120 minutes duration; for water flow above 1,750 gpm, use 180 minutes duration.

2 Separated from adjacent structures by at least 150 ft (46 m) on all sides.

3 Adjacent storage or structures located within 10 ft (3 m) on all sides.

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Next, Table 15 is used to determine a separation factor for each exposed facility. Separation factors for each exposed facility are summed to yield a single separation factor. For total separation factors of 15 or greater, the required flow is that flow necessary for the isolated storage case (given in Table 14), increased by 75 percent (i.e., multiple required Isolated Storage flow by 1.75). For separation factors less than 15, the percentage increase is determined by multiplying this separation factor by the value 5.35. For example, a separation factor of 10 yields a percentage increase of 10 x 5.35 or approximately 54 percent. Therefore, multiply the isolated storage flow by 1.54 to determine the required flow.

5.2.2.6 Vehicle Parking Areas. A minimum fire flow rate of 500 gal. per minute (1,920 l/min) for a 2-hr duration shall be provided for all such facilities.

5.3 Pressure Required. The minimum desired residual pressure is 20 psi when the required fire flow is delivered. For Navy Projects, when hose streams are to be taken directly from fire hydrants without the use of a fire department pumper, the minimum residual pressure shall be 90 psi (620 KPa). For Army and Air Force projects, detailed analysis will be based on TM 5-813-5/AFM 88-10 series, Chapter 5, and TM 5-813-5/AFM 88-10, Chapter 6 and Appendix C, para. 1.7 of this handbook.

5.4 Duration of Water Supplies. The sources of the fire protection water supply shall be capable of furnishing the necessary flow and pressure for the duration indicated in the appropriate tables. Duration of water supplies for deluge sprinkler systems is 45 minutes.

5.5 Quantities of Water Required. The requirements for fire protection water storage are based on the assumption that there will be only one fire at a time. The quantity of water required is equal to the product of the fire protection water demand and the required duration. This quantity represents fire protection requirements only, and must be available at all times. Water supply for domestic, industrial, and other demands must be added to these requirements in order to determine the total amount of water that is necessary at a facility. If the public water system supplying a facility is reliable, provides a minimum of two connections, each providing at least 50 percent of required capacity, has adequate capacity and pressure to meet water requirements and continuous reserve storage capacity at least equal to the required fire protection water storage, then no separate water storage facility is normally required.

NOTE: Refer to TM 5-813-5/AFM 88-10 for Army and Air Force Projects.

5.5.1 Total Storage Capacity. The total stored supply for fire protection purposes shall be sufficient to meet the maximum required fire flow demand. For combined storage for domestic, industrial and fire protection use, refer to the following engineering manuals. For Army and Air Force Projects, refer to TM-813-4/AFM 88-10, Vol. 4. For Navy Projects, refer to NAVFAC DM-5.7, Water Supply Systems.

5.5.2 Reduction in Storage Capacity. In computing the fire protection storage requirement, a reduction in storage capacity is acceptable if an adequate replenishment source is available. Factors that must be evaluated

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Table 15
Distance vs. Separation Factors for Facilities

SEPARATION DISTANCE (In Ft)	SEPARATION FACTOR
0 - 10	5
11 - 25	4
26 - 50	3
51 - 100	2
101 - 150	1
over 150 ft	

include the reliability of the makeup facility, its sustained flow capacity, its method of operation (automatic or manual), and flow limitations imposed by the capacity of treatment operations.

5.5.3 Replenishment of Storage. The water storage must be self-replenishing. It shall reach required volume during normal consumption within 48 hours, and within 24 hrs by curtailing normal consumption.

5.6 Sources of Supply

5.6.1 Primary Water Supplies. Primary water supplies shall consist of one or a combination of the following:

a) Two connections to a public water system (one connection is ample for a small activity, such as a Reserve Training Facility).
b) Elevated tanks or reservoirs.
c) Multiple pumps with adequate suction supply.

5.6.2 Secondary Water Supplies. Where public water supply is inadequate or unreliable, a secondary supply is required. Secondary supply shall be gravity tank, pressure tank, booster pumps taking suction from adequate capacity main, or fire pumps taking suction from adequate source.

5.7 Pumps

5.7.1 Requirements. Pumps for fire protection shall have adequate capacity with reliable power and water supply. This equipment shall conform to requirements of NFPA 20, Centrifugal Fire Pumps. Fire pumps, prime movers, and all other equipment including automatic accessories shall be listed by UL or approved by FM.

5.7.2 Pump Type. A fire pump may be either a horizontal or vertical shaft centrifugal pump or a vertical shaft turbine pump, whichever is most

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economical and appropriate for the intended use. A horizontal centrifugal pump in either the horizontal or vertical position will not be used where suction lift is required. A vertical shaft turbine pump will be used for suction lift.

5.7.3 Pump Starting Arrangement. Fire pumps must be arranged to start automatically except that they may be arranged for manual starting when other available water supply sources are capable of providing the demands for all automatic sprinkler and special fire suppression systems simultaneously with the domestic industrial demands. All fire pumps shall include manual shutdown feature.

5.7.4 Pump Drive. When electric power is economically available from a reliable single power source (average forced down time, excluding scheduled repairs, does not exceed 2 hours per year) or from two independent sources in accordance with NFPA 20, pumps will be electric-driven only. When such electric power supplies are not available, fire pumps will be diesel engine-driven. Spark ignited internal combustion engines will not be used to drive fire pumps.

5.8 Distribution Systems

5.8.1 Mains. The distribution system shall be sized to accommodate fire flows plus domestic and industrial or flushing demands that cannot be restricted during fires. Distribution mains shall be looped to provide at least 50 percent of the fire flow in case of a single break. Distribution systems shall be designed in accordance with TM 5-813-5/AFM 88-10, Vol. 5, for Army and Air Force Projects, and NAVFAC DM 5.7 and NFPA 24, Installation of Private Fire Service Mains and their Appurtenances, for Navy Projects.

5.8.2 Valves. Control valves shall be provided in each source of water supply (such as tanks and pumps). A sufficient number of sectional valves shall be provided so that not more than a combined total of five hydrants and sprinkler systems, or not more than three sprinkler systems shall be out of service due to a single break. Control valves shall be either post-indicating or outside-stem-and-yoke types. Sectional valves may be key-operated type. All new valves shall be right-hand valves.

5.8.2.1 Drawings. Drawings shall be provided showing all control and sectional valve locations, valve sizes, and number of turns. All left-hand valves shall be clearly indicated on drawings.

5.8.3 Hydrants. Fire hydrants shall be UL listed or FM approved and shall have two 2-1/2-in. (64 mm) hose outlets and one 4-in. (102 mm) or 4-1/2-in. (114 mm) suction connection with National Standard Firehose Threads in accordance with NFPA 24, and NFPA 1963, Firehose Connections. Wet-barrel or California-type hydrants are preferable in areas where there is no danger of freezing. If local municipal departments use nonstandard connections, adapters shall be made and supplied to engine companies that respond to station fires.

NOTE: The one 4-in. (102 mm) suction connection only applies to Navy projects.

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5.8.3.1 Installation Requirements. All hydrants shall be installed adjacent to paved areas, not closer than 3 ft (1 m) and not farther than 7 ft (2 m) from the roadway shoulder or curb line, where they will be accessible to fire department apparatus. Hydrants shall be installed with not less than 6 in. (152 mm) connection to the supply main, and valved at the connection. Barrels shall be long enough to permit at least 18 in. (457 mm) clearance between the center of the 4-1/2 in. (114 mm) pumper connection and grade. The ground shall be graded so that any surface drainage is away from the hydrant. Installation shall be in accordance with NFPA 24, except as modified in this handbook. Pumper connection should be perpendicular to the street to allow straight line connection to pumper.

NOTE: For Army and Air Force projects, hydrants shall be installed not closer than 6 ft (1.8 m) and not further than 7 ft (2.1 m) from paved surfaces.

5.8.3.2 Spacing Requirements. A sufficient number of hydrants shall be provided so that the hose stream demand can be met without more than 750 gal. per minute (2,820 l/min) from any single hydrant. Hydrants shall also be spaced in accordance with the following requirements:

NOTE: For Army and Air Force projects, hydrant distribution shall follow the guidance of Table 16.

a) All parts of buildings shall be reached by hose lays of not over 350 ft (110 m) with consideration given to accessibility and obstructions.

b) Hydrants protecting warehouses shall be spaced a maximum of 400 ft (120 m) apart.

c) Hydrants protecting aircraft hangars shall be located at 300 ft (90 m) maximum intervals, and there must be at least one hydrant at each corner of the hangar. Where required hydrants may obstruct aircraft movements, they shall be flush type.

d) Hydrants protecting POL storage and distribution facilities shall be spaced at 300 ft maximum intervals.

5.8.3.3 Color Coding of Fire Hydrants. Fire hydrants shall be color coded in accordance with the latest edition of NFPA's, Fire Protection Handbook.

5.8.4 Pressure Regulating Valves. Pressure regulating valves are restricted in use on fire protection water systems by NFPA 24. Where essential, they shall be installed on individual services rather than on the main piping. Where Pressure-Regulating Valves (PRVs) are provided in mains supplying systems or portions of systems with fire hydrants, automatic sprinkler systems, or other installed fire protection, the following features shall be provided to safeguard against failures and to facilitate maintenance:

a) Control valves on each side of PRVs,

b) Bypasses around PRVs, and

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Table 16
Hydrant Distribution

REQUIRED FLOW (gpm)	AVERAGE AREA/HYDRANT (Ft^2)
1,000 or less	160,000
1,500	150,000
2,000	140,000
2,500	130,000
3,000	120,000
3,500	110,000
4,000	100,000
4,500	95,000
5,000	90,000
5,500	85,000
6,000	80,000
6,500	75,000
7,000	70,000
7,500	65,000
8,000	60,000
8,500	57,500
9,000	55,000
10,000	50,000
11,000	45,000
12,000	40,000

c) Pressure relief valves on low pressure side of PRVs.

5.8.5 Water Level Controls. Manual controls, double-acting altitude valves, or other automatic devices shall be used to maintain the water level in elevated storage tanks. Altitude valves shall be arranged with bypasses.

5.8.6 Meters. Where meters are installed on water distribution systems they shall be FM approved or UL listed fire flow meters.

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Section 6: FIRE EXTINGUISHING SYSTEMS

6.1 Automated Sprinkler Systems. This para. cross references para. 15-70 of Air Force Regulation (AFR) 88-15, Criteria and Standards for Air Force Construction.

6.1.1 Characteristics. Properly engineered and installed automatic sprinkler systems are designed to detect the presence of fire, activate both local and remote (fire department) alarms, and distribute water in sufficient quantity to either control or extinguish it.

6.1.2 Application. The following factors shall be considered when determining the necessity and justification for automatic sprinkler installation in a given facility:

a) Life, safety, and occupancy hazard risks

b) Exterior fire exposures presented by the facility

c) Strategic or tactical importance of the facility or supplies housed therein

d) The quantity of supplies or facilities involved relative to available supplies or facilities of the same type

e) Ability to replace the facility or its contents

f) Inherent fire hazards of the facility

g) Monetary value

h) Size of the facility in relation to available fire fighting resources, and

i) Permanence of the facility.

6.1.3 Protected Occupancies. Automatic sprinkler protection shall be provided in the following buildings:

a) Hospital buildings and other medical facilities except that small, one-story hospitals and clinics which are of Type II, one-hr protected, noncombustible construction may be nonsprinkled.

b) Buildings of combustible construction used for the following:

(1) Confinement of military prisoners.

(2) Bachelor quarters (barracks) and quarters with fire areas in excess of 5,000 ft^2 (450 m^2) per floor or which are three stories or more in height.

c) Warehouses, sheds (including transit and pier sheds), and similar storage-type buildings which are:

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(1) mission essential,

(2) severe fire hazard, and

(3) of high monetary value.

d) Technical, industrial, and commercial type buildings, including hangars, shops, and laboratories used for production, repair, experimental testing, electronics overhaul facilities, or other processes, services, or equipment, which are:

(1) mission essential,

(2) severe fire hazard,

(3) of high monetary value.

e) Buildings of combustible construction with fire areas in excess of 8,000 ft^2 (740 m^2) or that are three stories or more in height.

f) Commissaries and exchanges of combustible construction having fire areas in excess of 8,000 ft^2.

g) Windowless or underground structures when the combustibility of the contents or life-safety features warrant the protection.

h) Storage and other areas of buildings of noncombustible or fire-resistive construction where moderate or large volumes of combustible materials are stored or which constitute fire or explosion hazards. These areas shall also be separated from the areas without sprinklers by fire partitions having at least a 1-hr fire-resistance rating and with fire doors of comparable rating at the openings.

i) Extra hazard occupancies as defined in Appendix B.

j) Buildings in which significant amounts of critical equipment or material are exposed by combustible construction, storage, or hazardous operations. Critical equipment or material includes training devices (such as flight, navigational, or gunnery). Sprinklered areas shall be separated from unsprinklered areas by partitions having a minimum fire-resistant rating of 1 hr.

k) Buildings used for the manufacture, testing, or processing of explosives, propellants, pyrotechnics, signaling equipment, and similar items.

k) Buildings five stories or higher, measured from the lowest grade level.

l) Aircraft maintenance hangars (refer to para. 4.4.6).

m) Aircraft acoustical enclosures (refer to para. 4.4.7).

6.1.4 Design Requirements. All sprinkler systems shall use equipment and devices listed by UL or approved by FM. Sprinkler systems for light-,

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ordinary-, and extra-hazard occupancies shall follow applicable criteria set forth in NFPA 13, Sprinkler Systems, and NFPA 16, Foam Water Sprinkler and Spray Systems. The sprinkler water requirements are stipulated in para. 5.1 of this handbook. Sprinkler system layouts should not normally be included in contract drawings. Shop drawings shall be submitted by the sprinkler system contractor.

6.1.4.1 Hydraulic Calculations. The required discharge densities and areas of discharge operation are given in Section 5. Calculations shall follow the format of NFPA 13. Pipe friction losses and equivalent lengths of pipe for fittings and valves shall be in accordance with NFPA 13. The following shall also apply:

a) Piping Layout. For discharge densities of 0.20 gpm/ft^2 (8 l/min/m^2) or less, the distance between lines shall not exceed that permitted by NFPA 13 for ordinary hazard occupancies. For discharge densities of more than 0.20 gpm/ft^2 (8 l/min/m^2), the distance between lines shall not exceed that permitted by NFPA 13 for extra hazard occupancies.

b) Spacing of Sprinklers on Lines. The distance between sprinklers on lines shall be approximately the same as the distance between lines.

6.1.4.2 Sprinkler Coverage. Where sprinkler protection is justified, the sprinklers shall provide 100 percent coverage. An exception is small area spot protection in buildings having a low order of hazard overall. Partitions of at least 1-hr fire resistive construction are required between sprinklered and unsprinklered areas.

6.1.4.3 Connections to Exterior Fire Reporting Systems. Where sprinklers are located at facilities with station or base fire reporting systems, the sprinkler systems shall be electrically connected to the fire reporting system for transmission of sprinkler water flow alarms.

6.1.4.4 Strainers. Where water conditions warrant, or systems involve high velocity flows, riser strainers shall be installed.

6.2 Water Spray Systems

6.2.1 Requirements. Design requirements for water spray systems shall conform to NFPA 15, Water Spray Fixed Systems For Fire Protection.

6.3 Foam Systems

6.3 1 Requirements. The NFPA Fire Protection Handbook and the FM Loss Prevention Data Sheets contain data and information concerning the installation and arrangement of foam systems for various types of Class B (flammable liquids) hazards. Foam installations shall be in accordance with NFPA 11, Foam Extinguishing Systems, and NFPA 11A, High Expansion Foam Systems, and NFPA 11B, Synthetic Foam and Combined Agent Systems.

6.3.2 Design. The installation shall be designed for use against fires in gasoline storage even though fuel oil may actually be stored, so that change in the type of fuel stored does not require any major change in design.

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6.4 Standpipe Systems

6.4.1 Requirements. Installation of 2-1/2-in. (64 mm) standpipe systems shall be in accordance with NFPA 14, Standpipe and Hose Systems. Standpipes with 2-1/2-in. hose outlets (without hose) shall be provided in stair towers of buildings greater than three stories in height. Provide Class 1 or Class 11 standpipe systems in all windowless and underground structures.

6.5 Dry Chemical Systems

6.5.1 Application. Fixed dry chemical systems are suitable for protection of certain types of special occupancies and facilities, such as cooking exhaust systems, dip tanks, and other flammable liquid operations.

6.5.2 Requirements and Limitations. Requirements for installation and operation are contained in NFPA 17, Dry Chemical Extinguishing Systems. Dry chemicals shall not be used in electronics occupancies.

6.6 Carbon Dioxide Systems

6.6.1 Application. Carbon dioxide systems are normally effective against flammable liquid (Class B) and electrical (Class C) fires.

6.6.2 Design Requirements. Carbon dioxide systems shall conform to NFPA 12, Carbon Dioxide Extinguishing Systems. Where applicable, UL listed or FM approved components shall be used. These systems are not authorized in new Air Force construction except in special applications as approved by HQ USAF/LEEE.

6.7 Halon 1301 Systems

6.7.1 Application. This para. cross references para. 15-70 of AFR 88-15, Criteria and Standards for Air Force Construction. Installations shall be limited to those applications described in Section 4.

6.7.2 Design Requirements. Halon 1301 systems shall conform to NFPA 12A, Halogenated Fire Extinguishing Agent Systems, Halon 1301. Where applicable, UL listed or FM approved components shall be used.

6.8 Portable Extinguishers. This para. cross references para. 15-67 of Air Force Regulation (AFR) 88-15, Criteria and Standards for Air Force Construction.

6.8.1 Requirements. Portable extinguishers shall be provided in accordance with NFPA 10, Portable Fire Extinguishers. Portable fire extinguishers are not required in Government controlled family housing.

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Section 7: ALARM AND DETECTION SYSTEMS

7.1 Fire Reporting Systems

7.1.1 Application. Fire reporting systems shall be installed in occupied areas and buildings as a means for automatically and manually reporting fires to station or base fire department headquarters or other central alarm location as required to implement fire fighting operations and emergency action. Such systems may be of either telegraphic, radio, or supervised conductor types. Extensions of fire reporting systems will require consideration of compatibility with existing equipment. Fire reporting systems shall not be provided at isolated small areas, ammunition and ordnance storage, and similar restricted areas where personnel are not generally expected to be present. Fire reporting systems are also not required in family housing areas that are remote from main built-up area of the activity or station.

7.1.2 Requirements. Alarm reporting systems shall conform to NFPA 1221, Public Fire Service Communications, or NFPA 72D, Proprietary Protection Signaling Systems and shall provide the following where applicable:

a) Transmission of coded signals to fire department headquarters and other central locations,
b) Permanent record of alarm signal, time, and date,
c) Automatic supervision of alarm initiating circuits,
d) Automatic conditioning for transmission of signals under line fault conditions or single ground, single open, or both, and
e) Automatic testing of radio signaling devices.

7.1.3 System Design. Basic fire alarm systems shall be designed in accordance with applicable requirements of NFPA Standards 72D, or NFPA 1221.

Contract drawings shall indicate locations of boxes, panels, registers, auxiliary signaling devices, bells, horns, location lights, batteries, and other system components. Wiring diagrams shall be schematic to permit variations peculiar to equipment of different manufacturers. Specifications shall explicitly describe functional, quality, and test requirements.

7.2 Evacuation Fire Alarm Systems

7.2.1 Requirements. Fire evacuation alarm systems shall be installed in accordance with NFPA Standards 72A, Local Protection Signaling Systems, and 72D, Proprietary Protective Signaling Systems. Evacuation fire alarm systems shall be provided in:

a) All buildings as required by NFPA 101, Life Safety Code. These systems shall be arranged to transmit signals to a central alarm location or fire department, where possible,

b) Barracks, UOPH, UEPH and similar sleeping facilities, and

c) All buildings which require that the automatic fire detection system be connected to a central alarm location or fire department.

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7.3 Fire Detection Systems

7.3.1 Requirements. All fire detection installations shall conform to the applicable provisions of NFPA Standards Nos. 72A, Local Protective Signaling Systems; 72D, Proprietary Protective Signaling Systems; 72E, Automatic Fire Detectors; and 74, Household Fire Warning Equipment. Detection systems shall be arranged to alert building occupants of an alarm condition and to transmit a signal to a central alarm location or fire department. Where a station fire department does not exist, alarms shall be transmitted to a constantly attended location. Single station detectors as required in one- and two-family housing do not require transmission. Battery-powered smoke detectors are not approved for new housing.

7.3.1.1 General. Detection systems are required for the following:

a) All areas requiring detection systems by NFPA standards.

b) Bachelor officers and enlisted personnel housing (UOPH and UEPH). Refer to para. 4.2.1 for further information.

c) Family housing (refer to para. 4.2.2).

d) Electronic equipment installations (refer to para. 4.3.5).

e) Ordnance facilities (refer to para. 4.4.2).

7.3.1.2 Smoke Detection and Destratification Fans. The area of protection for smoke detection devices permitted by NFPA 72E shall be reduced by 50 percent in areas where destratification fans are used. This restriction does not apply to thermal or flame detection devices or to residential occupancies.

7.3.1.3 Minimum Level of Protection. For Air Force projects, unless automatic suppression systems are required by this handbook or other guidance, complete automatic fire detection systems and portable fire extinguishers constitute the minimum level of acceptable fire protection for all new construction. Smoke detectors in return air plenums do not meet this minimum level.

7.3.1.4 Scope of Protected Spaces. Areas protected with automatic fire detection systems shall include occupied spaces, unoccupied spaces and attics.

7.3.1.5 Graphic Annunciators. For Air Force projects, when a system is divided into four or more zones, a graphic annunciator constructed of engraved phenolic or metal material shall be installed at a location determined by the host fire department.

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APPENDIX A

CLASSIFICATION OF OCCUPANCIES

1 Classification of Occupancies. Occupancies shall be categorized by one of seven major occupancy classification groups specified in the Uniform Building Code. Classifications are Groups A, B, E, H, I, M or R.

1.1 Group A Occupancies. This classification consists of the following:

a) Division 1 - Any assembly building with a stage and an occupant load of 1000 or more in the building.

b) Division 2 - Any building or portion of a building having an assembly room with an occupant load of less than 1,000 and a stage.

c) Division 2.1 - Any building or portion of a building having an assembly room with an occupant load of 300 or more without a stage, including such buildings used for educational purposes and not classed as a Group E or Group B, Division 2 Occupancy.

d) Division 3 - Any building or portion of a building having an assembly room with an occupant load of less than 300 without a stage, including such buildings used for educational purposes and not classed as a Group E or Group B, Division 2 Occupancy.

e) Division 4 - Stadiums, reviewing stands and amusement park structures not included within other Group A Occupancies.

1.2 Group B Occupancies. This classification consists of the following:

a) Division 1 - Gasoline service stations, garages where no repair work is done except exchange of parts and maintenance requiring no open flame, welding or use of highly flammable liquids.

b) Division 2 - Drinking and dining establishments having an occupant load of less than 50; wholesale and retail stores, office buildings, printing plants, base police and fire stations; workshops using materials not highly flammable or combustible; storage and sales rooms for combustible goods, paint stores without bulk handling; buildings or portions of buildings having rooms used for educational purposes with less than 50 occupants in any room.

c) Division 3 - Open parking garages, heliports, and aircraft hangars where no repair work is done except exchange of parts and maintenance requiring no open flame, welding or the use of highly flammable liquids.

d) Division 4 - Ice plants, power plants, pumping plants, cold storage and creameries, factories and workshops using noncombustible and nonexplosive materials; storage and sales rooms of noncombustible and nonexplosive materials.

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1.3 Group E Occupancies. This classification consists of the following:

a) Division 1 - Any building used for educational purposes through the 12th grade by 50 or more persons for more than 12 hours per week or 4 hours in any one day.

b) Division 2 - Any building used for educational purposes through the 12th grade by less than 50 persons for more than 12 hours per week or 4 hours in any one day.

c) Division 3 - Any building used for day-care purposes for more than six children.

1.4 Group H Occupancies. This classification consists of the following:

a) Division 1 - Storage, handling, use or sale of hazardous and highly flammable or explosive materials other than flammable liquids.

b) Division 2 - Storage, handling, use or sale of Classes I, II and III-A liquids; dry cleaning plants using Class I, II or III-A liquids; paint stores with bulk handling; paint shops and spray-painting rooms and shops.

c) Division 3 - Woodworking establishments, planing mills, box factories, shops, factories or warehouses where loose combustible fibers or dust are manufactured, processed, generated or stored; and pin-refinishing rooms.

d) Division 4 - Repair garages not classified as Group B, Division 1.

e) Division 5 - Aircraft repair hangars.

f) Special Provisions. In buildings used for educational purposes, vocational shops, laboratories and similar areas need not be classified as Group H Occupancies, provided:

(1) Such areas are separated from classrooms other than the class-room directly related to the use by not less than a 1-hr fire-resistive occupancy separation, and

(2) Such areas are separated from each other by not less than a 1-hr fire-resistive occupancy separation unless the uses are determined to be compatible.

(3) In any room in a Division 1, 2, or 3 Occupancy in which volatile, flammable liquids or hazardous materials are stored or used, energy consuming equipment shall not be used unless such equipment has been listed specifically for the hazardous atmosphere that may develop.

(4) In Division 4 Occupancies, devices which generate a spark or glow capable of igniting gasoline vapors shall not be installed or used within 18 in. (457 mm) of the floor.

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(5) The requirements of the UBC Fire Code shall be met when applicable.

Where an approved spray booth constructed as specified in the UBC Fire Code is installed, such booth need not be separated from other Group H or Group B Occupancies.

1.5 Group I Occupancies. This classification consists of the following:

a) Division 1 - Nurseries for the full-time care of children under the age of 6 (each accommodating more than five persons); hospitals, sanitariums, nursing homes with nonambulatory patients and similar buildings (each accommodating more than five persons).

b) Division 2 - Nursing homes for ambulatory patients, homes for children 6 years of age or over (each accommodating more than five persons).

c) Division 3 - Mental hospitals, mental sanitariums, jails, prisons, reformatories and buildings where personal liberties of inmates are similarly restrained.

1.6 Group M Occupancies. This classification consists of the following:

a) Division 1 - Private garages, carports, sheds and agricultural buildings.

b) Division 2 - Tanks and towers.

1.7 Group R Occupancies. This classification consists of the following:

a) Division 1 - Hotels and apartment houses.

b) Division 2 - Not used.

c) Division 3 - Dwellings and lodging houses.

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APPENDIX B

OCCUPANCY HAZARD CLASSIFICATION

1 Classification of Occupancies. The principal occupancy classifications are light hazard, ordinary hazard and extra hazard and are dependent upon the degree of fire hazard present. Typical examples are listed under each classification. The basic hazard classification of an occupancy does not, in all instances, categorically define the fire hazard present in all areas of that occupancy. If more hazardous processes or areas exist within a given occupancy, they shall be protected in accordance with the fire protection requirements pertaining to the hazard classification of that area. The classification for unlisted occupancies shall be determined from the definitions or by comparison with one of the listed occupancies.

1.1 Light Hazard Occupancies. Occupancies or portions of other occupancies where the quantity and combustibility of contents are low and fires with relatively low rates of heat release are expected. Small, scattered amounts of flammable liquids in closed containers are allowable in quantities not exceeding 5 gal (20 l). The following are examples of Light Hazard Occupancies:

Churches and Chapels
Clinics (Dental, Outpatient)
Data Processing Areas
Disciplinary Barracks
Dispensaries
Dwellings
Drill Halls
Gymnasiums
Hospitals
Mess Areas
Offices
Photographic Processing Areas
Schools
Barracks

1.2 Ordinary Hazard Group 1 Occupancies. Occupancies or portions of other occupancies where combustibility is low, quantity of combustibles is moderate, stock piles of combustibles do not exceed 8 ft (2.5 m), and fires with moderate rates of heat release are expected. Modest, scattered amounts of flammable liquids, in closed containers, are allowable in quantities up to about 20 gal (75 l). The following are examples of Ordinary Hazard Group 1 Occupancies:

Armories
Bowling Alleys
Clubs (Officer, Enlisted Personnel, etc.)
Commissaries
Exchanges
Forge Shops
Machine Shops
Printing Shops (Using inks having flash points above 110° F [44° C])
Sheet Metal Shops
Shipfitting Shops
Kitchens
Small Stores
Theatres and Auditoriums
Welding Shops

1.3 Ordinary Hazard Group 2 Occupancies. Occupancies or portions of other occupancies where quantity and combustibility of contents is moderate, stock piles do not exceed 12 ft (4 m), and fires with moderate rate of heat release are expected. Moderate, scattered amounts of flammable liquids in closed containers are allowable in quantities up to about 50 gal (200 l).

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Small amounts of flammable liquids may be exposed as required by normal operations. The following are examples of Ordinary Hazard Group 2 Occupancies:

Air Rework Facilities
Boiler Rooms
Electrical Maintenance Shops
Engine and Generator Rooms
Laboratories
Laundries
Refrigeration and Air Compressor Rooms
Switchgear Rooms

1.4 Ordinary Hazard Group 3 Occupancies. Occupancies or portions of other occupancies where quantity or combustibility of contents is high and fires of high rate of heat release are expected. Moderate, scattered amounts of flammable liquids in closed containers are allowable in quantities up to about 50 gal (200 l). Small amounts of flammable liquids may be exposed as required by normal operations. The following are examples of Ordinary Hazard Group 3 Occupancies:

Piers and Wharves
Vehicle Repair Garages
Ordnance Plants (without exposed powder)
Woodworking Plants
Warehouses (moderate to high combustibility and not high piled or high rack storage)

1.5 Extra Hazard Occupancies. Occupancies or portions of other occupancies where quantity and combustibility of contents are very high, or where flammable liquids, dust, lint, or other materials are present, introducing the probability of explosion and rapidly developing fires with high rates of heat release. The following are examples of Extra Hazard Occupancies:

a) Flammable and Combustible Liquids (not high piled or high rack storage)

b) Warehouses (extra high combustibility of contents)

1.6 Special Occupancies. Special occupancies are facilities or areas which cannot be assigned a specific classification because of special protection requirements. Refer to the appropriate chapters or paragraphs of this handbook for fire protection guidance. This classification includes, but is not restricted to, the following occupancies:

Aircraft Hangars
Engine Test Cells
Missile Assembly
Ordnance Plants (exposed powder areas)
Rubber Tire Storage
Warehouses (high piled or high rack storage)
Foam Rubber or Plastic Storage

NOTE: Refer to National Fire Protection Association Fire Codes for occupancies not listed.

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APPENDIX C

PROCEDURE FOR DETERMINING

BUILDING FIRE FLOW DEMAND IN UNSPRINKLERED FACILITIES

1 General. The following steps are to be used in determining the required fire flow demand in unsprinklered facilities. Deviation from these steps may be required depending upon the particular circumstances at the subject facility.

1.1 Step One - Establish Building Parameters. Building parameters shall be established as follows:

a) Determine the type of construction of the subject facility. Construction can be determined from architectural plans or by inspecting building structural components.

b) Determine the number of stories in the subject facility. This handbook differentiates between single story buildings and multiple story buildings only.

c) Determine the hazard classification of the subject facility occupancies as defined in Appendix B.

d) Determine the floor area of each story of the subject facility.

e) Determine the distance from the subject facility to each exposed facility. This distance is the shortest distance measured from the subject facility wall to the exposed facility, and rounded to the nearest foot. This distance is to be measured for each exposed facility.

1.2 Step Two - Develop Basic Fire Flow Rate. Using the parameters obtained in step one, choose the appropriate figure to use in developing the basic fire flow rate. The figures are grouped into three sets based on type of construction:

a) Figures C-1 and C-1A contain curves to be used with Type I (fire-resistive) and Type II (fire-resistive/protected noncombustible) construction

b) Figures C-2 and C-2A contain curves to be used with Type II (1 Hr/N rating unprotected noncombustible construction

c) Figures C-3 and C-3A contain curves to be used with Types III, IV, V (ordinary, heavy timber, wood frame) construction.

Determine the appropriate curve on the figure which is related to subject facility height (single story or multiple story) and hazard classification of the subject facility occupancies (light-, ordinary-, or extra-hazard).

NOTE: There are two figures for each type of construction curve--each with a different square footage and gpm range.

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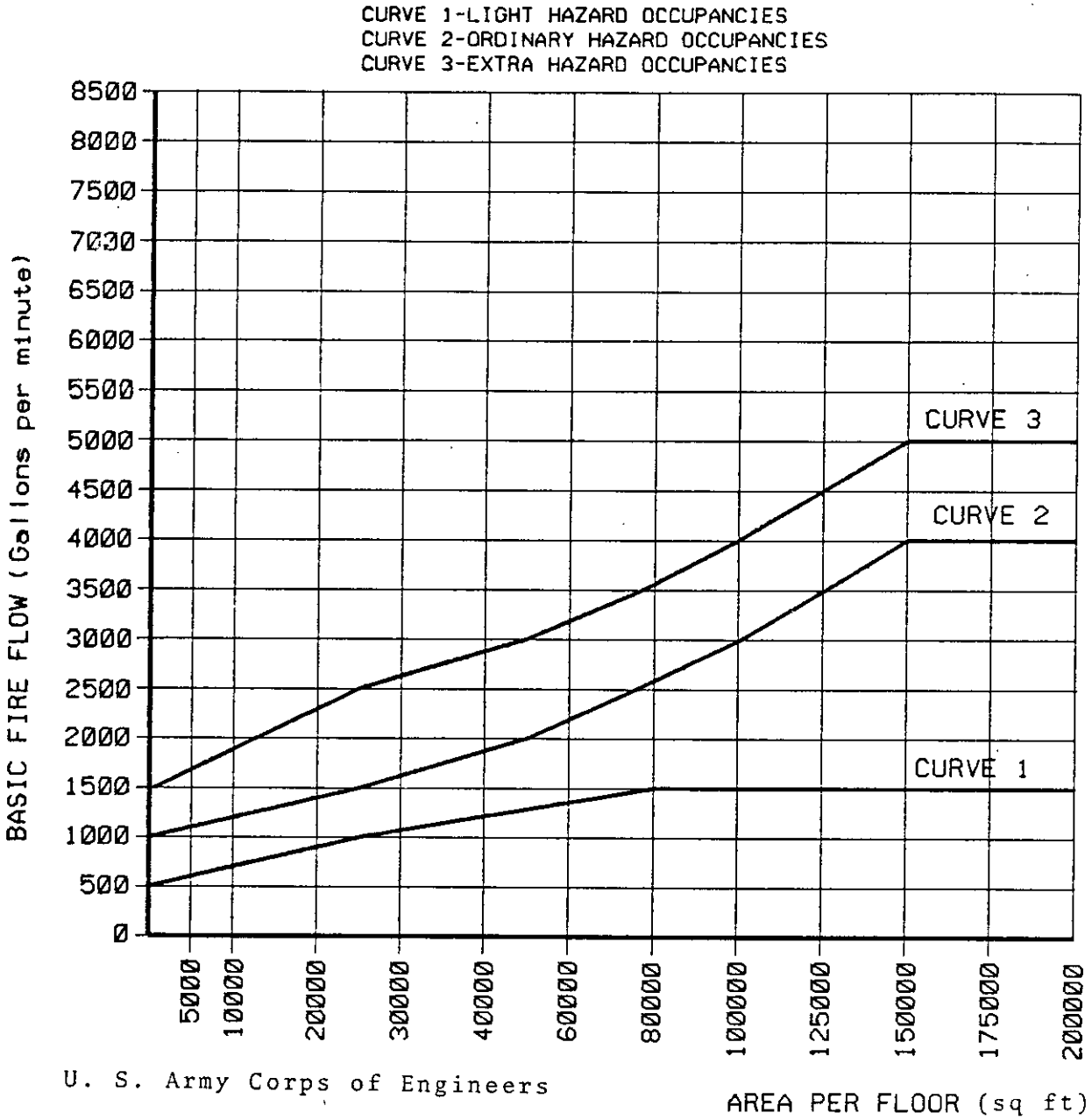


Figure C-1
Type I (Fire-Resistive) and Type II (Fire-Resistive/Protected Noncombustible) Construction Curves

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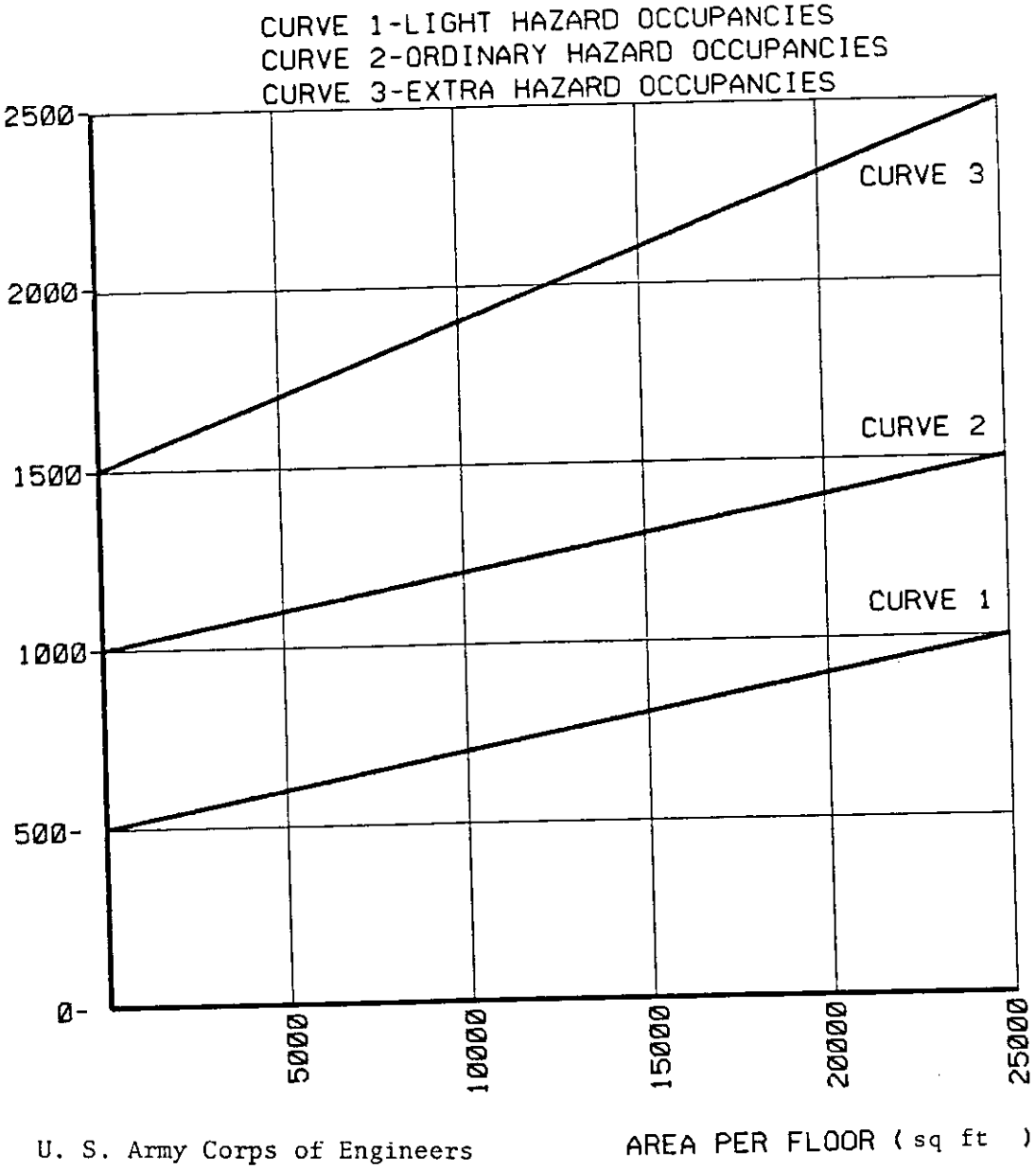


Figure C-1A
Type I (Fire-Resistive) and Type II (Fire-Resistive/Protected Noncombustible) Construction Curves

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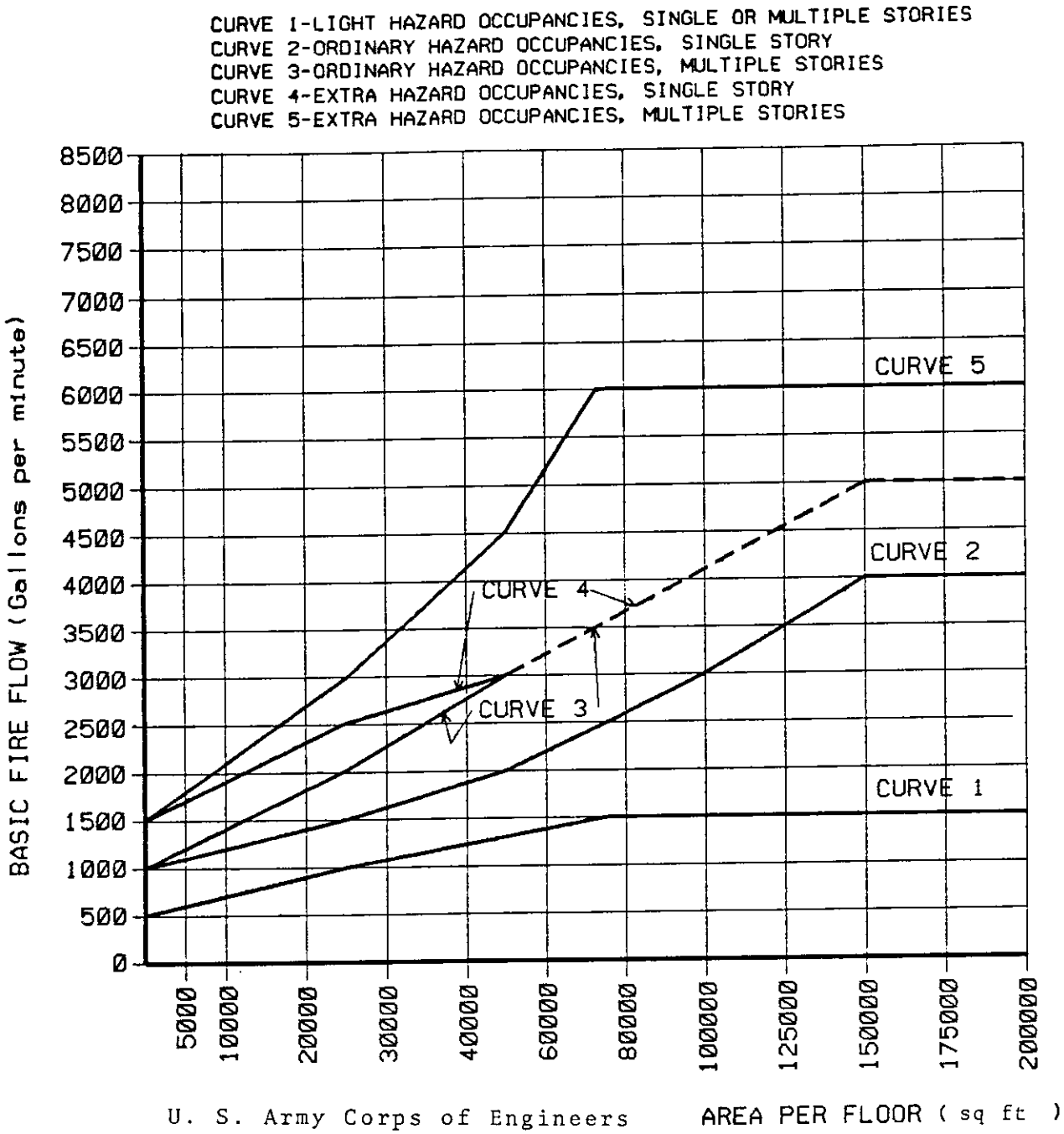


Figure C-2
Type II (1 H/N Rating/Unprotected Noncumbustible)
Construction Curves

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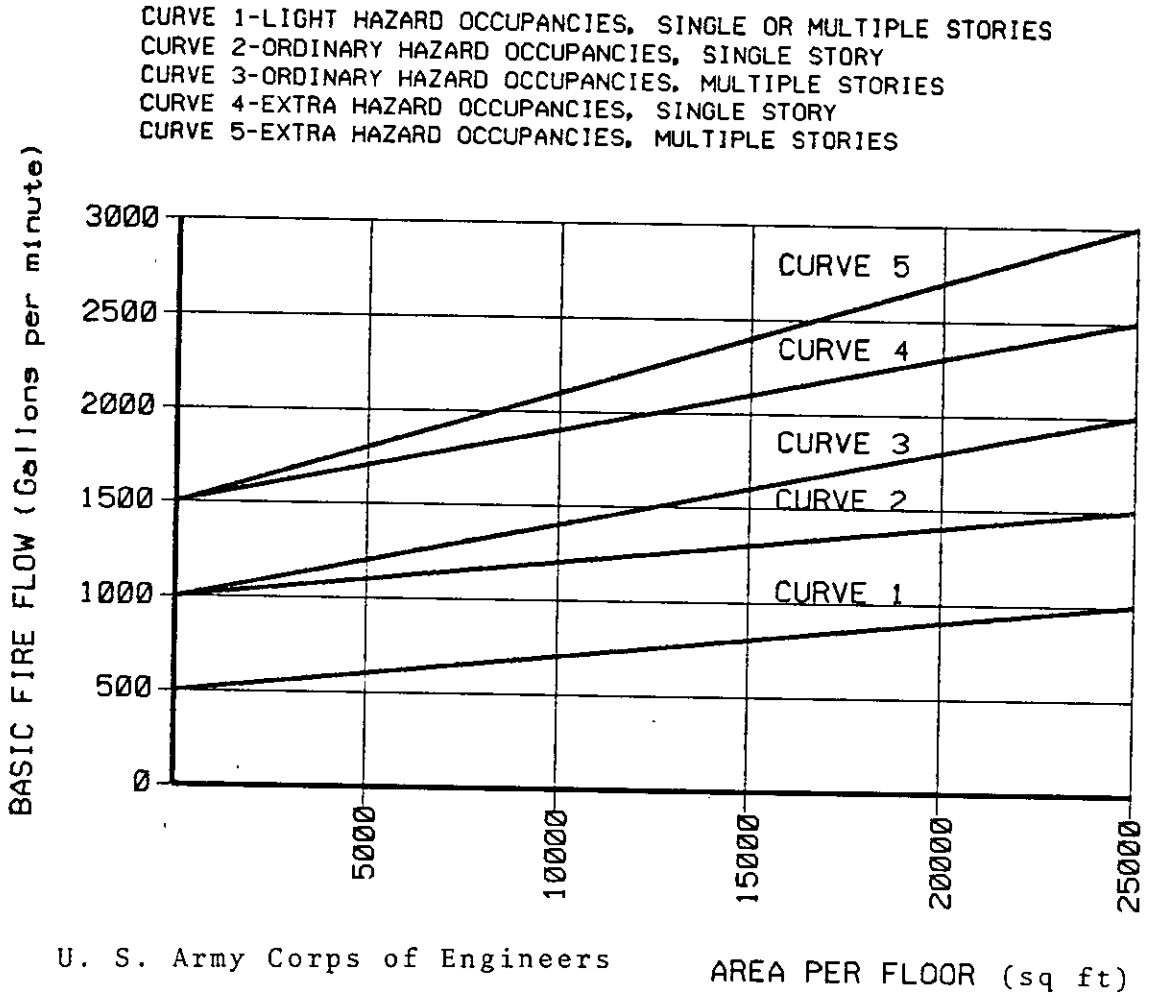


Figure C-2A
Type II (1 Hr/N Rating/Unprotected Noncumbustible)
Construction Curves

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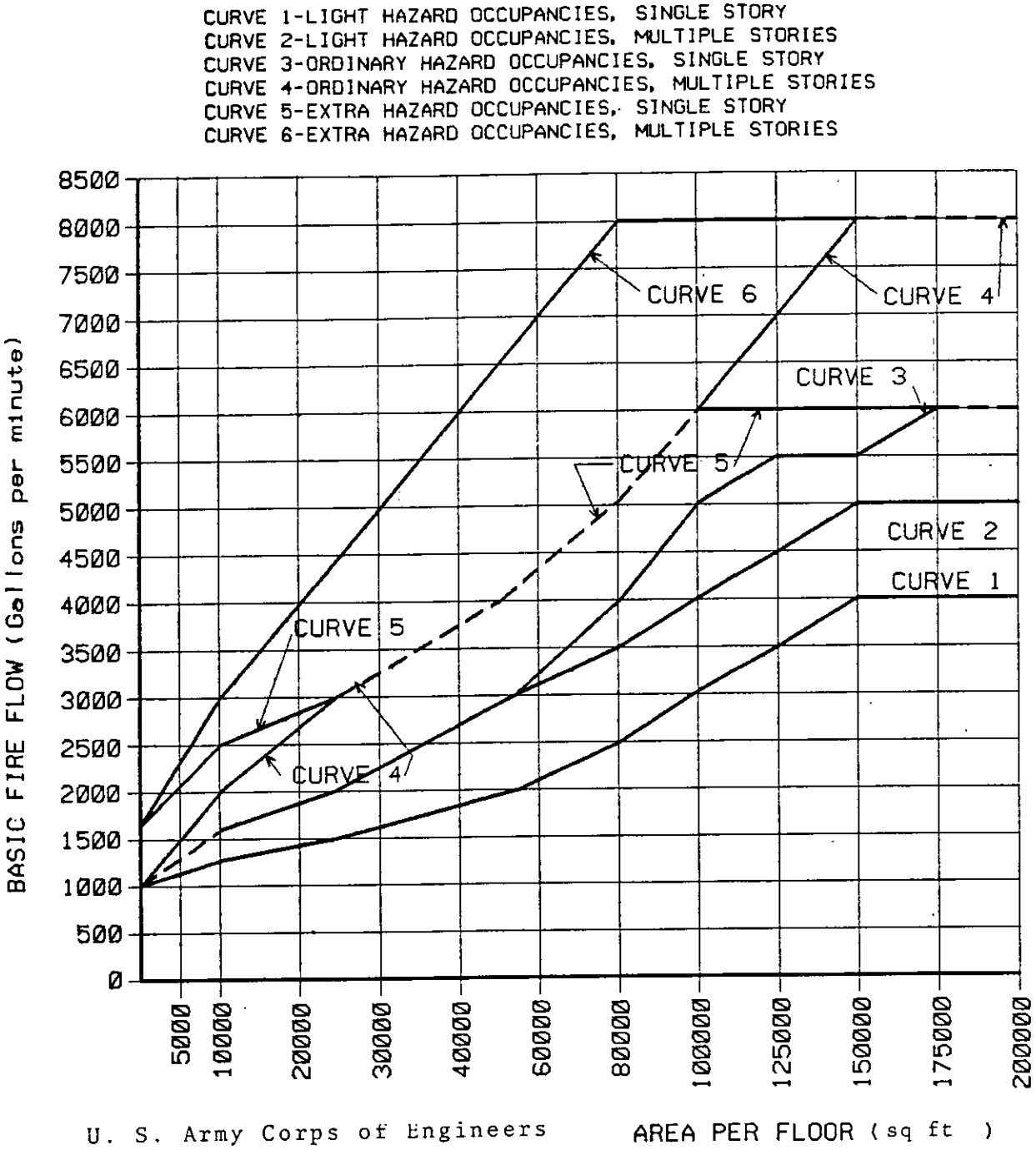


Figure C-3
Types III, IV, and V (Combustible) Construction Curves

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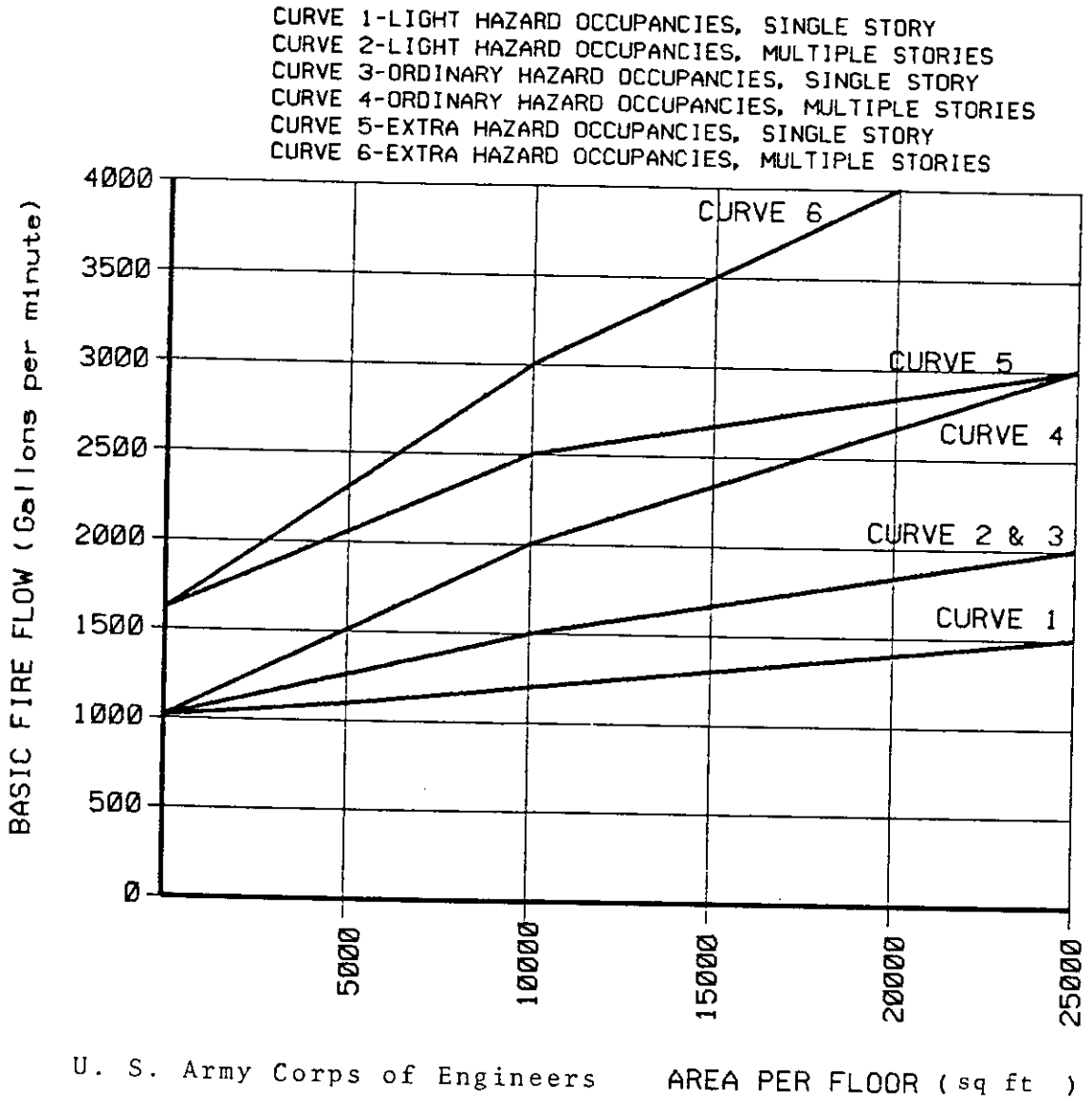


Figure C-3A
Type III, IV, and V (Combustible) Construction Curves

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1.3 Step Three - Determine Basic Fire Flow Rate. Determine the basic fire flow rate from the appropriate curve, using the area of the largest floor developed in para. 1.1.d.

1.4 Step Four - Exposure Protection. Determine the increase needed for exposure protection.

a) Determine the separation factor from Table 15 for each exposed facility, using the distance developed in para. 1.1.a.

b) Add the separation factors for all exposed facilities.

c) Determine the exposure factor from Figure C-4, using the sum of the separation factors calculated in para. 1.4.b. The maximum value of the exposure factor is 1.75, even though the separation factor total may be greater than 15.

d) Determine the required fire flow rate by multiplying the basic fire flow rate by the exposure factor, rounding to the nearest 100 gpm (379.5 lpm).

1.6 Step Five - Fire Flow Duration. Determine the required fire flow duration from Table C-1 using the required fire flow rate developed in para. 1.4, d.

1.7 Step Six - Determine Residual Pressure. Determine the minimum required residual pressure. The minimum desired residual pressure is 20 psi (137.9 kPa) when the required fire flow is delivered. The distribution system pressure shall be designed according to the criteria in TM-813-4; AFM 88-10 Vol. 4; and TM 5-813-5/AFM 88-10 Vol. 5 for Army and Air Force projects. New water supply, storage, and distribution system will be designed to provide the fire demand with the 20 psi residual pressure. When new facilities are provided fire flow from existing systems, fire flow tests will be run to determine the capability of the system to meet the desired 20 psi residual pressure. All fire flow tests and calculations to determine system capacity shall be used in accordance with the procedure in NFPA Standard No. 291. If the required flow is available with 20 psi residual pressure, or can be obtained with extensions to the existing system at the construction site, the design will be based on the desired 20 psi residual pressure. If the existing system cannot meet the requirement, a minimum residual pressure of 10 psi (68.95 kPa) will be provided. Improvement of existing facilities for te sole purpose of meeting the 20-psi residual requirement is not authorized. When improvement to the existing distribution network is required to meet the 10-psi minimum residual requirement, the system will be improved to meet the 20-psi residual requirement.

1.8 Step Seven - Additional Water Requirements. Determine the need for additional water flow rate, duration, or pressure requirements created by use of a partial automatic sprinkler system, or special fire suppression systems in the subject facility.

a) Determine partial automatic sprinkler system or special fire suppression system water requirements.

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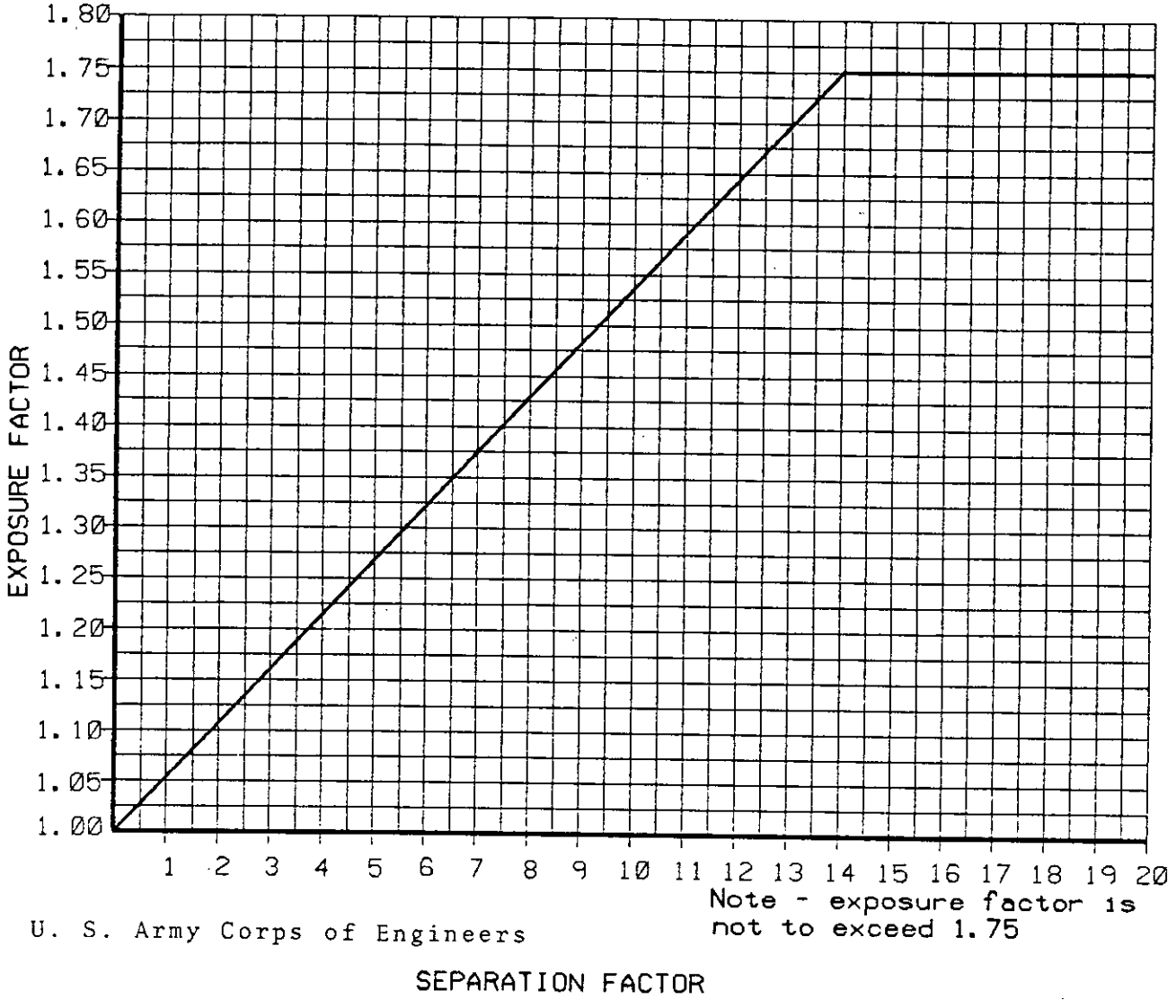


Figure C-4
Exposure Factor Curve

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Table C-1
Required Fire Flow vs. Duration

REQUIRED FIRE FLOW (gpm)	MINIMUM DURATION (hr)
1500 or less	2
1750 - 3500	3
3750 or over	4

Source: U.S. Army Corps of Engineers

b) Determine if increases are needed to the required fire flow rate, fire flow duration, or minimum residual pressure.

(1) If the partial automatic sprinkler or special fire suppression system flow rate, residual pressure and duration are less than those of the required fire flow demand, no adjustment is necessary.

(2) If the partial automatic sprinkler or special fire suppression system flow rate and duration requirements are less than those of the required fire flow demand, but the system residual pressure requirement is greater than that required fire flow demand, increase the pressure for the portion of the fire flow demand needed to operate the system. The remainder of the required fire flow demand may be supplied at the established residual pressure or at a higher pressure.

(3) If the partial automatic sprinkler or special fire suppression system flow rate requirement is greater than the required flow rate, the system flow rate requirement is to be used as the fire flow rate.

(4) If the partial automatic sprinkler or special fire suppression system fire flow duration requirement is greater than the fire flow duration requirement of the required fire flow demand, the system duration is to be used as the duration of the required fire flow demand.

1.8 Step Eight - Sprinklered Fire Flow Demand. Determine the required fire flow demand for a building protected by complete automatic sprinkler system, using Table 10. If special fire suppression systems are provided in addition to the sprinkler system, add the demands for te special systems to the sprinkler system demand unless otherwise directed.

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REFERENCES
(Publications containing criteria cited in this handbook)

ANSI Codes available from American National Standards Institute (ANSI), New York, N.Y. 10018.

ANSI A17.1b	Safety Code for Elevators, Dumbwaiters, Escalators, and Moving Walks.
ANSI-C2	National Electrical Safety Code.

API Publications Standards available from American Petroleum Institute (API), Washington, D.C. 20037.

ASME Code available from American Society of Mechanical Engineers (ASME), New York, N.Y. 10017.

ASME Boiler and Pressure Vessel Code.

ASTM Standards available from American Society for Testing and Materials (ASTM), Philadelphia, PA 19103.

ASTM D-92	Flash and Fire Points by Cleveland Open Cup, Test for
ASTM E-84	Test Method for Surface Burning Characteristics of Building Materials.
ASTM E-119	Standard Methods of Fire Tests of Building Construction and Materials.
ASTM E-648	Critical Radiant Flux of Floor Covering Systems Using a Radiant Flux Heat Source.
ASTM E-814	Fire Tests of Through-Penetration Fire Stops.

Armed Services Explosives Safety Board Publications available from U.S. Government Printing Office, Washington, DC 20402.

Explosives Safety Manual.
Safety Manual for Siting, Constructing and Equipping Pier and Wharf.
Facilities for Handling Explosives.
Structures to Resist the Effects of Accidental Explosions.

Army Corps of Engineers Publications available from Department of the Army, Office Chief Engineer DAEN-ASP, Forrestal Building, Washington, D.C. 20314.

EM385-1-1	General Safety Requirements.

Army Corps of Engineers Guide Specification available from Department of the Army, Publications Depot, Alexandria, VA 22304.

CEGS-08701	Hardware Prison-Locking Devices.

Department of Defense Publications available from Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

MIL-HDBK-1190	Facility Planning and Design Guide.
DOD 5154.4S	DOD Ammunition and Explosives Safety Standards.

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Department Of Labor, Occupational Safety and Health Administration, Code of Federal Regulations (CFR).

CFR 40 Series	Protection of Environment
CFR 40-264	Regulations for Owners and Operators of Hazardous Waste Facilities
CFR 40-265	Interim Station Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
CFR 40-270	EPA Administered Permit Program: The Hazardous Waste Permit Program

Federal Test Method Standards available from U.S. Government Printing Office, Washington, D.C. 20402.

No. 372	Test for Critical Radiant Flux of Carpet Flooring Systems (Flooring Radiant Panel Test).

Factory Mutual Publications available from Factory Mutual Engineering Corporation, Norwood, MA 02062.

Approval Guide.
Loss Prevention Data Sheets.

International Conference of Building Officials Publication available from International Conference of Building Officials, Whittier, CA 90601.

Uniform Building Code.

NFPA Publications available from National Fire Protection Association (NFPA), Boston, MA 02110.

NFPA 10	Portable Fire Extinguishers
NFPA 11	Foam Extinguishing Systems
NFPA 11A	High Expansion Foam Systems
NFPA 11B	Synthetic Foam and Combined Agent Systems
NFPA 12	Carbon Dioxide Extinguishing Systems
NFPA 12A	Halogenated Fire Extinguishing Agent Systems, Halon 1301
NFPA 13	Installation of Sprinkler Systems
NFPA 14	Standpipe and Hose Systems
NFPA 15	Water Spray Fixed Systems for Fire Protection
NFPA 16	Foam Water Sprinkler and Spray Systems
NFPA 17	Dry Chemical Extinguishing Systems
NFPA 20	Centrifugal Fire Pumps
NFPA 24	Private Fire Service Mains and Their Appurtenances
NFPA 30	Flammable and Combustible Liquids Code
NFPA 31	Installation of Oil Burning Equipment
NFPA 33	Spray Application Using Flammable and Combustible Materials
NFPA 34	Dipping and Coating Processes Using Flammable or Combustible Liquids
NFPA 37	Stationary Combustion Engines and Gas Turbines
NFPA 43A	Storage of Liquid and Solid Oxidizing Materials

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NFPA 46	Storage of Forest Products
NFPA 48	Storage, Handling and Processing of Magnesium
NFPA 49	Hazardous Chemical Data
NFPA 50	Bulk Oxygen Systems at Consumer Sites
NFPA 50A	Gaseous Hydrogen Systems at Consumer Sites
NFPA 50B	Liquefied Hydrogen Systems at Consumer Sites
NFPA 51	Oxygen-Fuel Gas Systems for Welding and Cutting
NFPA 54	National Fuel Gas Code
NFPA 56A	Inhalation Anesthetics
NFPA 56B	Respiratory Therapy
NFPA 56C	Laboratories in Health Related Institutions
NFPA 56D	Hyperbaric Facilities
NFPA 56E	Hypobaric Facilities
NFPA 56F	Nonflammable Medical Gas Systems
NFPA 58	Storage and Handling of Liquefied Petroleum Gases
NFPA 59A	Liquefied Natural Gas
NFPA 65	Aluminum, Processing and Finishing
NFPA 68	Explosion Venting Guide
NFPA 69	Explosion Prevention Systems
NFPA 70	National Electrical Code
NFPA 72A	Local Protection Signaling Systems
NFPA 72D	Proprietary Protection Signaling Systems
NFPA 72E	Automatic Fire Detectors
NFPA 74	Household Warning Equipment
NFPA 75	Computer/Data Processing Equipment
NFPA 76A	Essential Electrical Systems for Health Care Facilities
NFPA 76C	Safe Use of High Frequency Electricity in Health Care Facilities
NFPA 80	Fire Doors and Windows
NFPA 85	Oil and Natural Gas-Fired Single Burner Boiler-Furnaces
NFPA 85B	Natural Gas-Fired Multiple Burner Boiler-Furnace
NFPA 85D	Prevention of Furnace Explosions in Fuel Oil-Fired Multiple Burner Boiler-Furnaces
NFPA 85E	Prevention of Furnace Explosions in PulverizedCoal-Fire Multiple Burner Boiler-Furnaces
NFPA 85F	Installation and Operation of Pulverized Fuel Systems
NFPA 86A	Ovens and Furnaces
NFPA 86B	Industrial Furnaces
NFPA 86C	Industrial Furnaces Special Processing
NFPA 87	Piers and Wharves
NFPA 88B	Repair Garages
NFPA 89M	Clearance for Heat Producing Appliances
NFPA 90A	Standard for the Installation of Air Conditioning and Ventilating Systems
NFPA 90B	Standard for the Installation of Warm Air Heating and Air Conditioning Systems
NFPA 91	Standard for the Installation of Blower and Exhaust Systems for Dust, Stack and Vapor Removal or Conveying
NFPA 96	Removal of Smoke and Grease-Laden Vapors from Commercial Cooking Equipment
NFPA 101	Life Safety Code
NFPA 204	Smoke and Heat Venting Guide

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NFPA 211	Standard for Chimneys, Fireplaces, and Vents
NFPA 220	Building Construction, Standard Types
NFPA 231	Indoor General Storage
NFPA 231A	Recommended Safe Practices for Outdoor General Storage
NFPA 231C	Rack Storage of Materials
NFPA 231D	Storage of Rubber Tires
NFPA 232	Protection of Records
NFPA 241	Building Construction and Demolition Operations
NFPA 303	Marinas and Boatyards
NFPA 312	Fire Protection of Vessels During Construction, Repair, and Lay-up
NFPA 385	Tank Vehicles for Flammable and Combustible Liquids
NFPA 407	Aircraft Fuel Servicing
NFPA 409	Aircraft Hangars
NFPA 415	Aircraft Fueling Ramp Drainage
NFPA 481	Production, Processing, Handling and Storage of Titanium
NFPA 482M	Production, Processing, Handling and Storage of Zirconium
NFPA 490	Storage of Ammonium Nitrate
NFPA 492	Separation Distances of Ammonium Nitrate and Blasting Agents
NFPA 495	Explosive Materials
NFPA 501A	Mobile Homes Including Mobile Home Parks Requirements
NFPA 651	Manufacture of Aluminum or Magnesium Powder
NFPA 654	Prevention of Dust Explosion in Plastics Industry
NFPA 664	Prevention of Dust Explosions in Woodworking and Wood Floor Manufacturing Plants
NFPA 910	Libraries and Library Collections
NFPA 911	Museums and Museum Collections
NFPA 1963	Firehose Connections
NFPA 1221	Public Fire Service Communications

Handbook of Fire Protection.
National Fire Codes, Volumes 1-16 inclusive.

Navy Publications available from U.S. Naval Publications & Forms Center, Philadelphia, PA 19120.

NAVFAC DM-8	Fire Protection Engineering
NAVFAC DM-22	Petroleum Fuel Facilities
NAVFAC DM-25 Series	Waterfront Operational Activities.
NAVFAC DM-29 Series	Drydocking Facilities
NAVFAC DM-35	Family Housing
NAVFAC Inst. 11012.114	Bachelor Enlisted Quarters Design Guidance, New Construction Program
TS-07250	Sprayed Fire Protection for Navy Projects

Underwriters' Laboratories, Inc., Publications available from Underwriters' Laboratories, Inc., Northbrook, IL 60062.

UL Building Materials Directory.
UL Fire Resistance Directory.

UL790	Test Methods of Fire Resistance of Roof Covering Materials
UL992	UL Chamber Test

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U.S. Department of Commerce Publications available from U.S. Government Printing Office, Washington, D.C. 20402.

DOC FF1-7	Standard for the Surface Flammability of Carpet and Rugs
FPCA RP-1	Standard Practice for the Fire Protection of Essential Electronic Equipment Operations

U.S. Department of Housing and Urban Development Publications available from U.S. Government Printing Office, Washington, D.C. 20402.

HUD 4900.1	Minimum Property Standards, Single Family
HUD 4910.1	Minimum Property Standards, Multi-Family

CUSTODIANS:

ARMY-CE
NAVY-YD
AF-04

PREPARING ACTIVITY:
NAVY-YD
PROJECT NO. FACR-0244

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-HDBK-1008	FIRE PROTECTION FOR FACILITIES ENGINEERING DESIGN AND CONSTRUCTION

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ EFD/PWO ☐ AE ☐ USER ☐ CONTRACTOR ☐ MANUFACTURER ☐ OICC/ROICC ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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